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International Journal of Thermal Sciences

Date: April 2026

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Study on the influence of ring power ratio on heat flow behavior and solidification characteristics of the molten pool during ring-mode laser welding of aluminum alloy

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1. Introduction

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CRediT authorship contribution statement

Highlights

- The mechanism by which the ring power ratio determined the weld solidification mode was revealed.
- The increase in the ring power ratio decreased keyhole oscillation and enhance the probability of bubble escape.
- G and GR were reduced by the uniform energy distribution on the keyhole wall caused by the rise in the ring power ratio.
- The increased ring power ratios resulted in better mechanical properties.

Abstract

The ring power ratio (RPR), as a crucial control parameter in ring-mode laser welding (RMLW) of aluminum alloys, significantly affects the fluid flow and heat distribution within the molten pool, playing a decisive role in the solidification structure of the weld. This work first used experiments to quantitatively analyze the effects of various RPR on the weld surface morphology and porosity. Subsequently, a transient numerical model considering heat transfer and fluid flow in the molten pool was established to analyze the dynamic keyhole behavior, heat flow distribution in the molten pool, and solidification parameters at different locations under various RPR. The results of the study indicated that a higher RPR leads to a more regular flow pattern with fewer, longer-range flow paths, which effectively reduced the influence on the keyhole and mitigated keyhole oscillation, thus preventing the expansion and contraction of the molten pool. Furthermore, the wider molten pool area and the longer-range upward flow increased the probability of bubbles escaping. Increasing the RPR caused the laser energy to move towards the keyhole bottom, resulting in a more uniform energy distribution on the keyhole wall. Both the temperature gradient G and the cooling rate R were significantly decreased. The average grain size was smallest at the center of the cross-section and largest at the fusion boundary when the $RPR=0$ and $RPR=50\%$, respectively. Due to the smooth weld surface and minimal porosity, even without the smallest grain size, the increased RPR exhibited superior tensile strength. This work elucidated the mechanism by which the RPR regulated the molten pool dynamics and heat flow distribution, thereby controlling the solidification microstructure. This providing a dependable technique for customizing the microstructure of RMLW aluminum alloy joints and improving welding quality.

Keywords

Ring-mode laser welding; Ring power ratio; Dynamic keyhole; Molten pool heat flow distribution; Solidification characteristics

[Declaration of competing interest](#)

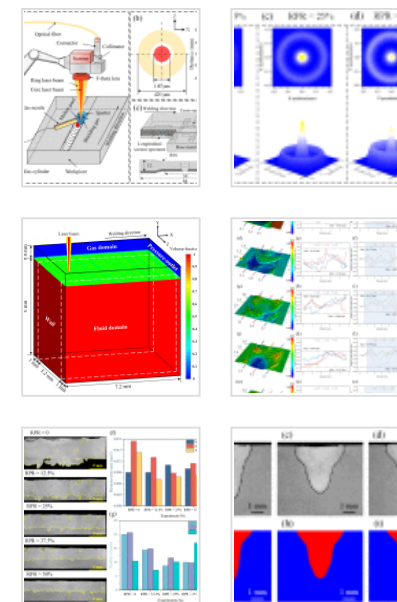
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1. Introduction

Aluminum alloys have become an indispensable structural material in high-end equipment manufacturing industries such as aerospace, new energy vehicles, and shipbuilding due to their excellent strength-to-weight ratio and processability. Achieving high-quality and efficient joining of aluminum alloys is one of the key technological challenges for promoting lightweight design in these industries [1,2]. Laser welding, as a highly efficient and precise joining method, is a key process in manufacturing lightweight aluminum alloy structures [3,4]. However, owing to the inherent properties of aluminum alloys, traditional single-mode Gaussian laser welding frequently encounters numerous challenges: high energy density leads to complex keyhole oscillation and molten pool flow patterns [5], unstable welding processes cause spatter and internal porosity defects [6], and non-uniform energy distribution and rapid cooling result in inhomogeneous solidification structures [7]. These problems severely restrict the improvement of joint performance and the reliability of the welding process. Currently, the primary technical strategies employed to overcome these issues include process parameter optimization [8,9], oscillating laser beams [10,11], energy field assistance [12,13], sub-atmospheric pressure environments [14,15], and hybrid welding techniques [16,17]. Han et al. [18] employed an oscillating beam and Zr in-situ alloying welding strategy to achieve ultrafine equiaxed grains throughout the full weld of medium-thick Al-Li alloy, thereby effectively mitigating uneven solidification structures and crack defects, and increasing the tensile strength of the joint by 66%. Xu et al. [19] used an oscillating magnetic field-assisted vacuum laser to achieve full-penetration welding of thick A5083 aluminum alloy. The effects of different environmental pressures, magnetic field strengths, and oscillation frequencies on weld quality were systematically studied. The results showed that high-quality welds could be obtained when the environmental pressure was 3–10kPa, the magnetic induction intensity was 60–80mT, and the magnetic field frequency was 300–

400Hz. Jiang et al. [20] studied the effect of oscillation frequency on the stability of a 10mm thick 6082-T6 alloy during oscillating Laser-MIG hybrid welding. The study showed that an oscillation frequency of 300–400Hz can maintain a balance between aspect ratio and porosity, and an increase in oscillation frequency significantly increased the proportion of equiaxed crystals in the weld center. Although there have been some positive results with these approaches, they frequently lead to increased system complexity and are also limited by the requirement for complex process parameter matching.

By precisely and independently controlling the core and ring energy distributions, ring-mode laser welding (RMLW), an advanced beam shaping technology, enables the precise adjustment of the welding thermal process. This provides a new approach to addressing the aforementioned issues [21,22]. Current research primarily focuses on macro-scale process optimization [23] and characterization of the final weld quality, including surface morphology [24], porosity [25], cracks [26], and mechanical strength [27]. Numerical simulation method can help overcome the limitations of experimental observations in accurately reproducing the real physical phenomena of welding processes. It can also monitor variables that are difficult to measure directly in experiments, thus enabling in-depth research into keyhole dynamics, heat transfer, and molten metal flow within the molten pool. Li et al. [28] revealed the mechanism for pore reduction using a 3D transient coupled model, finding that the ring beam stabilized the keyhole and molten pool, which reduced the pores generated at the gas-liquid interface. It also lowered the frequency of molten pool intersected keyhole, which decreased pores formed at the liquid-solid interface. Additionally, they elucidated the mechanism for suppressing spatter, established a quantitative model for preheating temperature and key process parameters, and derived the process parameters for achieving spatter-free welding [29]. Hu et al. [30] proposed a combined heat source model to analyze the dynamics of the molten pool and keyhole in RMLW. They discovered that the ring beam enhances the stirring effect of the molten pool by increasing its oscillation amplitude and prolongs the bubble escape time by reducing the oscillation frequency, thus achieving high-quality welds with minimal

porosity. Moore et al. [31] investigated a sequentially coupled heat transfer-Monte Carlo multiscale coupling model to investigate the temperature distribution and microstructure evolution during additive manufacturing under three beam modes. The simulation results showed that the ring beam altered the temperature amplitude and the direction of the temperature gradient, resulting in equiaxed grains at the weld center. The Bessel beam had the smallest molten pool and grain size. Pamarthi et al. [32] explored the impact of the ring beam on microstructure and cracking utilizing multi-physics CFD simulation. According to their research, increasing the ring beam diameter reduced melt flow and temperature gradients, thus minimizing cracking. Increasing the ring beam power promoted the formation of columnar crystals along the axial direction and coarsened the grain structure. Lee et al. [33] simulated the fluid flow characteristics of the molten pool during coaxial dual-beam laser welding of aluminum alloys by establishing a three-dimensional heat transfer and flow model. They quantitatively characterized the solidification mode induced by the additional ring beam based on solidification parameters. The simulation results showed that the ring beam reduced the surface temperature gradient of the molten pool, thereby reducing keyhole collapse and spatter. The increase in ring power increased the range of equiaxed grains.

The aforementioned studies have advanced our understanding of the mechanism of ring beam interaction, but they neglected the effects of transient pool dynamics and heat flow distribution under various ring power ratios (RPR) on the final solidification characteristics. The RPR, as a key control parameter, directly determines the spatial and temporal distribution of heat input, which fundamentally affects the keyhole dynamics and the heat-flow distribution within the molten pool, thereby determining the microstructure of the weld solidification. A thorough understanding of the relationship between the RPR, the heat flow behavior in the molten pool, and the solidification characteristics is crucial for optimizing the RMLW process of aluminum alloys, controlling microstructure, and producing high-performance joints. The absence of such a mechanistic understanding hinders the transition from empirical trial-and-error methods to intelligent, mechanism-driven optimization in RMLW.

This paper systematically investigated the effects of various RPR on keyhole dynamics, heat flow distribution in the molten pool, and the morphology and dimensions of the weld solidification structure. Based on process experiments, the surface morphology and porosity of weld seams under various RPR were analyzed. A numerical model was used to compare the dynamic keyhole behavior, heat transfer, fluid flow, and solidification parameters at different locations within the molten pool under various RPR. EBSD analysis was then employed to characterize the grain morphology and size of the weld, as well as to investigate its impact on the mechanical properties of the weld. This work provides valuable insights into the relationship between process parameters, heat flow, and microstructure in RMLW of aluminum alloys. It offers theoretical support for accurately managing the microstructure of the weld and enhancing welding quality.

2. Methodology

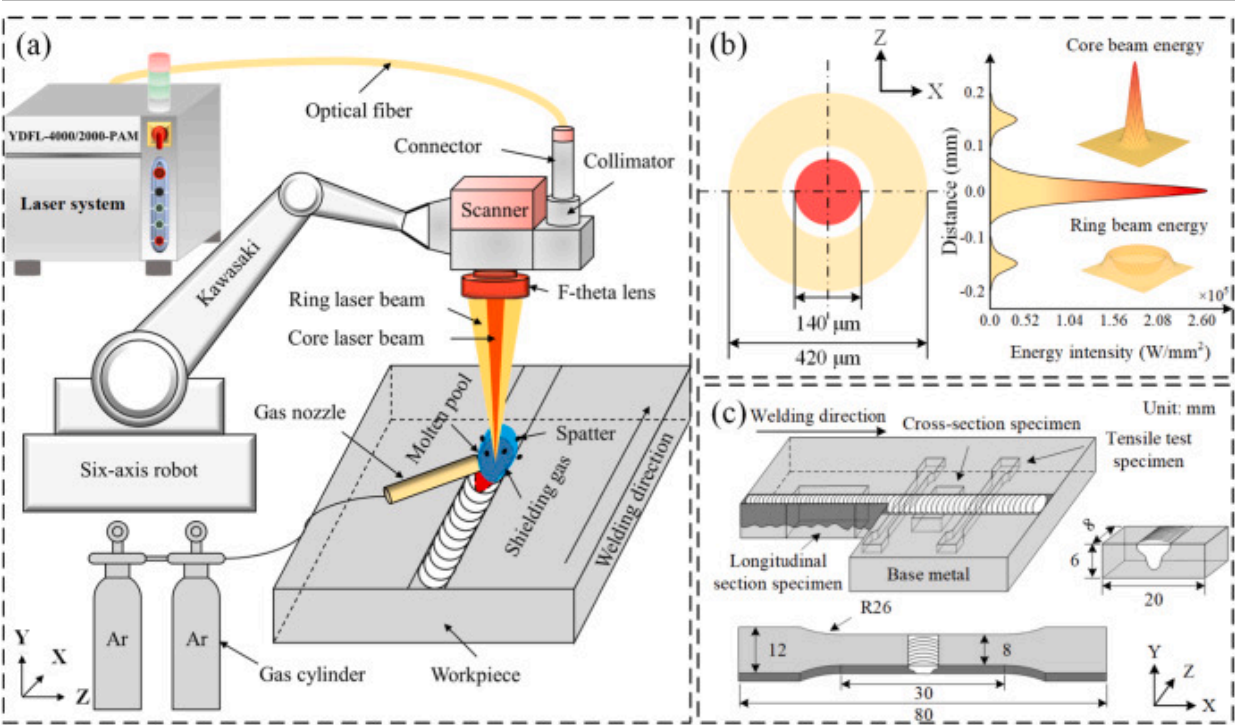
2.1. Experimental setup

In this study, 5A06-H112 aluminum alloy plates with a size of 100mm×100mm×6mm were used. Before welding, the surface of the area to be welded was polished along the welding direction using 80# sandpaper to remove the oxide film and create rough grooves parallel to the welding direction, thereby increasing the absorption of laser energy. Then, the surface contaminants were cleaned with acetone and dried for later use. The chemical composition of the base material is shown in [Table 1](#). The welding experimental device, as well as the spot size and energy distribution on the workpiece surface, are displayed in [Fig. 1\(a\)](#) and (b). A YDFL-4000/2000-PAM (FEIBO Co., Ltd., China) fiber laser with a maximum output power of 6kW was utilized. Its core diameter was 50μm, whereas its ring diameter was 150μm. The coherent laser parameters are listed in [Table 2](#). In order to study the influence of varied RPR, the following fixed parameters are used: total power of 4kW, welding speed of 20mm/s, shielding gas flow rate of 20L/min, and defocus of 0mm. [Table 3](#) shows the other main welding parameters. P_C and P_R represent core power and ring power, respectively. The energy density distribution at the

focal plane for different core beam and ring beam power configurations corresponding to Table 3 is shown in Fig. 2.

Table 1. The chemical composition of 5A06 aluminum alloy (wt.%).

Mg	Mn	Si	Zn	Zr	Fe	Cu	Ti	Al
6.2	0.69	0.21	0.19	0.2	0.14	0.1	0.05	the rest



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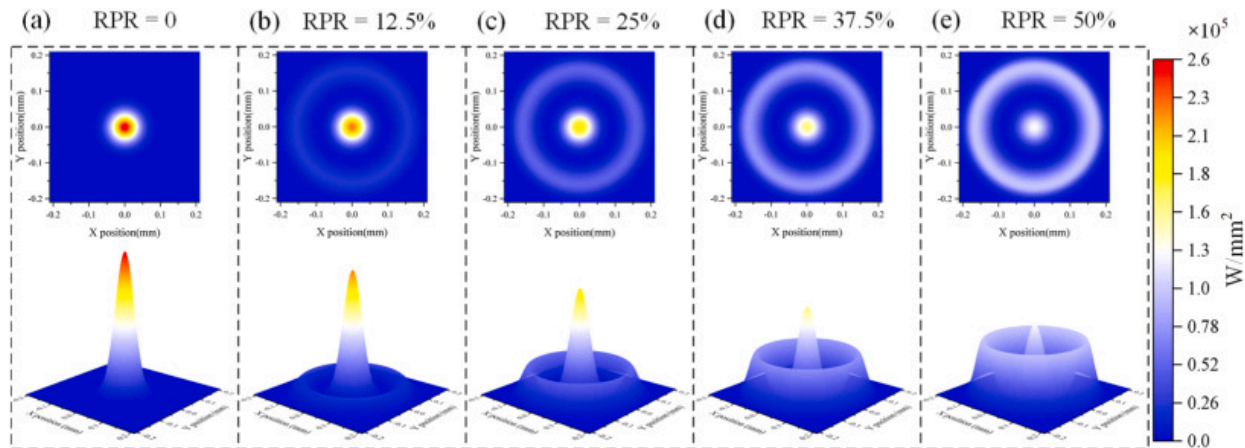
Fig. 1. (a) Schematic diagram of the laser welding experimental setup; (b) beam spot size and energy distribution on the workpiece surface; (c) schematic diagram of sampling location of test specimen.

Table 2. Specifications of the welding equipment.

Parameter	Core	Ring	Unit
Maximum power	4000	2000	W
Collimating length	150		mm
Focusing length	420		mm
Beam Product Parameter	<2.2	<7	mm·mrad
Optical fiber diameter	50	150	μm
Spot diameter on focus	140	420	μm
Wavelength	1.07–1.09		μm
Focal position	0	0	mm

Table 3. Experimental parameters of laser welding.

ID	Core power P_C/kW	Ring power P_R/kW	Ring power ratio RPR/%	Welding speed $v/\text{mm s}^{-1}$
P_{C4}	4.0	0	0	20
$P_{C3.5}P_{R0.5}$	3.5	0.5	12.5	20
$P_{C3}P_{R1}$	3.0	1.0	25	20
$P_{C2.5}P_{R1.5}$	2.5	1.5	37.5	20
$P_{C2}P_{R2}$	2.0	2.0	50	20



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Fig. 2. Energy density distribution of different core beam and ring beam power configurations.

The welded plates were electro spark machined to provide entire transverse and longitudinal weld cross sections. The sampling positions and sample sizes of the samples to be analyzed are shown in Fig. 1(c). The sample surface was ground with silicon carbide sandpaper ranging from 80# to 5000#, and then polished to a mirror finish using a 1 μm diamond suspension. The polished sample surface was then chemically etched for approximately 40s using Keller's reagent (1% HF+2.5% HNO₃+1.5% HCl+95% H₂O). The KEYENCE VK-X1000 three-dimensional stereomicroscope was used to observe the cross-sectional profile of the weld, the variation in weld penetration, and the distribution of porosity. ImageJ software was used to statistically analyze the pore size and calculate the porosity of the weld. The porosity was calculated as the percentage of the total pore area to the total weld area, expressed by Eq. (1). An OLYMPUS LEXT OLS5000 laser confocal microscope was used to quantify the weld surface height. The grain morphology was acquired using an Oxford Instruments Symmetry electron backscatter diffraction (EBSD) probe with an accelerating voltage of 20kV and a step size of 0.8 μm . The tensile test was

performed using a KY-100KNW universal tensile machine at a speed of 1 mm/min according to GB/T 228–2002. The preparation process for the tensile specimens was described in the author's previous study [34]. A TESCAN LYRA3 scanning electron microscope was used to observe the tensile fracture morphology, and an HXS-1000AC was used to measure the Vickers hardness of different areas of the weld. The test force was 50 gf, the holding time was 20s, and the adjacent indentation spacing outside the heat-affected zone was 0.2mm. The indentation spacing inside the heat-affected zone was appropriately reduced to obtain sufficient data.

$$p = \frac{\sum S_p}{S_w} \quad (1)$$

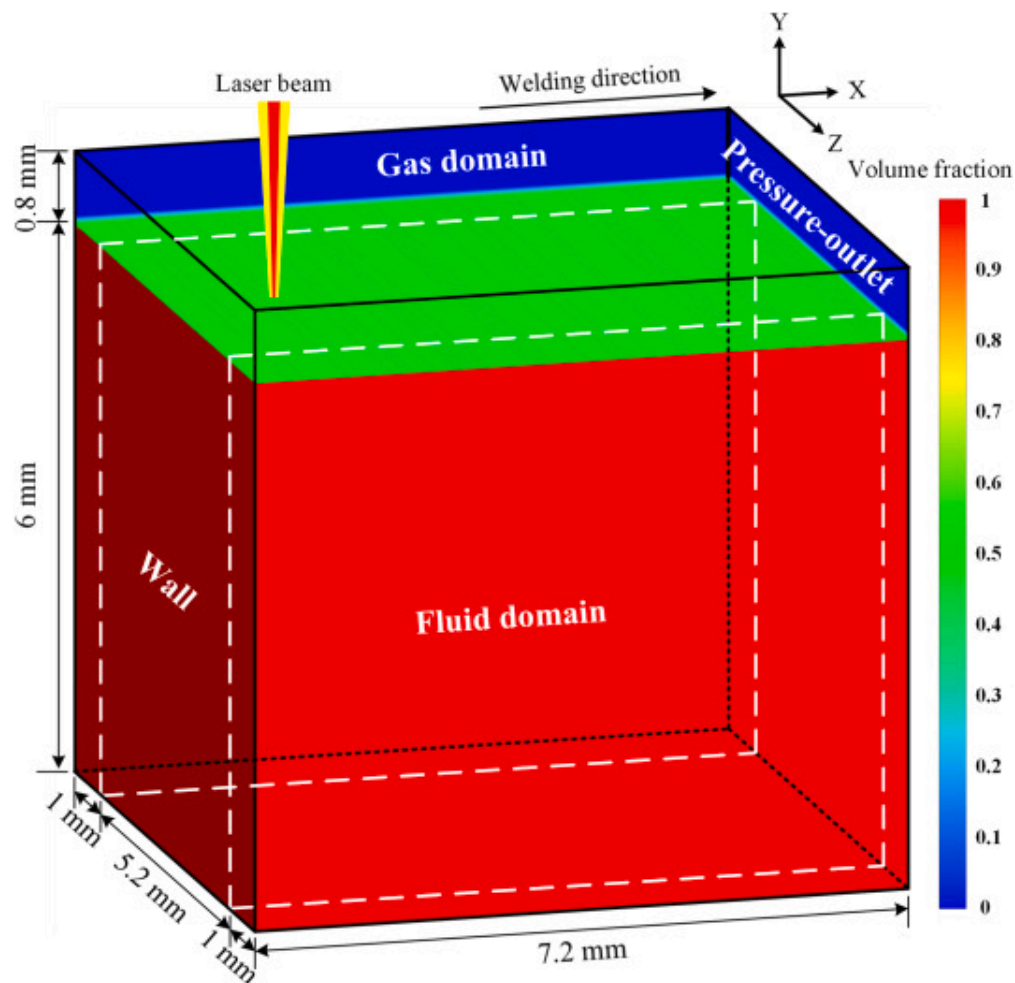
Where $\sum S_p$ represents the total area of pores, and S_w denotes the total area of the weld area.

2.2. Model implementation

In this paper, a three-dimensional transient multi-physics field model that considered physical phenomena such as multiple reflections, recoil pressure, surface tension, and vapor shear force was established to simulate the heat-flow distribution of the molten pool and the dynamic behavior of the keyhole during the RMLW process of aluminum alloy. Since the laser welding process in keyhole mode involves numerous complicated processes occurring simultaneously, we simplify the model for computational efficiency and make the following assumptions [35,36]: (1) The molten metal is considered to be a laminar, incompressible Newtonian fluid; (2) The material is isotropic and homogeneous, with the effect of the shielding gas is ignored; (3) The impact of the vapor plume is considered indirectly rather than directly.

The computational domain and boundary conditions of the simulation solution are depicted in Fig. 3. Since this paper involved two process conditions, partial penetration and full penetration, two different computational domains were created. The computational domain size for the partial penetration condition was 7.2mm×6.8mm×7.2mm. The domain consisted of two subdomains: a fluid domain with a

thickness of 6mm and a gas domain with a thickness of 0.8mm. The bottom and side surfaces of the fluid domain were considered as wall boundaries to account for thermal convection and radiation, and the top and side surfaces of the gas domain were designated as pressure-outlets. The entire domain was divided using a hexahedral grid, and two different mesh sizes were used to balance the computational efficiency: a 0.08mm uniform cube fine grid was applied to the range of 7.2mm×6.8mm×5.2mm, and a 0.2mm rectangular coarse grid was applied to the remaining area in the weld width direction, where only heat transfer occurs. The total number of cells is 573750. The computational domain size under full penetration conditions was 7.2mm×7.6mm×7.2mm, with an additional 0.8mm thick gas phase domain at the bottom of the fluid domain. The meshing scheme was consistent with that under partial penetration conditions. The total number of elements is 641,250. [Table 4](#) provides the thermophysical properties of aluminum alloy and argon gas used in the simulation calculation [[37,38](#)]. The calculation solution is based on the commercial software ANSYS FLUENT.



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Fig. 3. Schematic diagram of the computational domain used for simulation.

Table 4. Thermophysical parameters of 5A06 aluminum alloy and Argon [37,38].

Physical properties	5A06 Al	Argon
Density, (kg m^{-3})	2630	1.62

Physical properties	5A06 Al	Argon
Solidus temperature, (K)	833	
Liquidus temperature, (K)	905	
Specific heat at solidus, (J Kg ⁻¹ K ⁻¹)	940	520.6
Specific heat at liquidus, (J Kg ⁻¹ K ⁻¹)	1180	
Thermal conductivity at solidus, (W m ⁻¹ K ⁻¹)	143	0.0158
Thermal conductivity at liquidus, (W m ⁻¹ K ⁻¹)	84	
Boiling temperature, (K)	2720	
Latent heat of fusion, (J Kg ⁻¹)	3.87×10 ⁵	
Latent heat of evaporation, (J Kg ⁻¹)	1.08×10 ⁷	
Dynamic viscosity, (kg m ⁻¹ s)	0.0013	2.125×10 ⁻⁵
Thermal expansion coefficient	1.92×10 ⁻⁵	
Surface tension coefficient, (N m ⁻¹)	0.914	
Surface tension gradient, (N m ⁻¹ K ⁻¹)	-0.00035	
Convection heat transfer coefficient, (W K ⁻¹ m ⁻²)	15	
Thermal emissivity	0.08	

2.3. Governing equations

The conservation equations for mass, momentum, and energy are given follows:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 \quad (2)$$

$$\rho \left(\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{v} \right) = -\nabla P + \mu \nabla^2 \mathbf{v} - \mu C \frac{(1-f_L)^2}{f_L^3 + m} \mathbf{v} + \mathbf{F}_m \quad (3)$$

$$\frac{\partial(\rho h)}{\partial t} + \nabla \cdot (\rho h \mathbf{v}) = \nabla \cdot (\lambda \nabla T) + S_q \quad (4)$$

Where ρ , t , and $\mathbf{v}$ represent fluid density, time and velocity vectors respectively. P , μ , C denote pressure, dynamic viscosity, and mushy zone characteristics, respectively. f_L is the liquid fraction, m represents a very small number that prevents division by zero, and $\mathbf{F}_m$ is the force source term. In Eq. (4), h , λ and S_q represent enthalpy, thermal conductivity, and energy source terms, respectively.

The Coupled Level-Set and Volume of Fluid (CLSVOF) algorithm is used to handle the complicated topological evolution of the gas-liquid interface. The complementary properties of the two methods allow for more precise location of the free interface [39].

$$H = \frac{1}{2}(1 + \tanh(h_s \phi)), h_s = 0.85dx \quad (5)$$

The following convection equation is then solved to determine the exact location of the free surface of the molten pool.

$$\frac{\partial H}{\partial t} + \mathbf{v} \cdot \nabla H = 0 \quad (6)$$

Where H , ϕ and h_s are VOF function, LS function and interface thickness respectively. The material properties of the mixed fluid in each cell are considered as a mixture of two phases connected by volume fractions.

$$\varphi = \varphi_1 H_1 + \varphi_2 (1 - H_1) \quad (7)$$

Where φ_1 , φ_2 and H_1 represent the material properties of phase-1 and phase-2 and the fraction of phase-1, respectively.

2.4. Laser heat source model

The core and ring beams of ring-mode can be regarded as Gaussian distributions, and the heat source model consists of the following two parts:

$$Q = \frac{3Q_c}{\pi R_c^2} \exp\left(-\frac{3r^2}{R_c^2}\right) + \frac{3Q_r}{\pi \omega^2} \exp\left(-\frac{3(r-R_{r1})^2}{\omega^2}\right) \quad (8)$$

Where Q_c and Q_r represent the laser powers of the core and ring beams, respectively. R_c denote the core beams' spot radius in the focal plane. R_{r1} and ω indicate the inner diameter of the spot of the ring beam at the focal plane and the half bandwidth of the ring beam, respectively. The reflection of the laser beam on the keyhole wall satisfies the following relationship. $\mathbf{x}_{i+1}$, $\mathbf{x}_i$ and $\mathbf{n}$ are the direction of the reverse beam, the direction of the incident beam, and the normal direction of the free interface, respectively.

$$\mathbf{x}_{i+1} = \mathbf{x}_i - 2(\mathbf{x}_i \cdot \mathbf{n}) \mathbf{n} \quad (9)$$

2.5. Main driving forces

The main driving forces in the RMLW process include recoil pressure $\mathbf{P}_r$, surface tension $\mathbf{P}_\sigma$, vapor shear stress $\mathbf{P}_{vap}$ and $\mathbf{P}_{sta}$, and buoyancy $\mathbf{F}_\beta$ [40]:

$$F_m = AP_0 \exp\left(L_m \frac{T-T_v}{RTT_v}\right) + \kappa[\sigma_0 + A_\sigma(T - T_m)] + \frac{8\rho_g(v_g \sin \theta)^2}{R_e} + \frac{\rho_g(v_g \cos \theta)^2}{2} + \rho g \beta(T - T_{ref}) \quad (10)$$

Where A is a numerical coefficient, P_0 is the ambient pressure. L_m , T_v and R denote the latent heat of evaporation, boiling temperature and gas constant, respectively. κ , σ_0 , A_σ and T_m represent the curvature of the free surface, surface tension coefficient, temperature coefficient of surface tension, and melting point, respectively. ρ_g denotes the vapor density, and R_e is the Reynolds number. β corresponds to the thermal expansion coefficient, and T_{ref} is the reference temperature of 300K.

$v_g \sin \theta$ and $v_g \cos \theta$ are the two components of the vapor velocity v_g in the direction parallel and perpendicular to the keyhole wall, respectively, and are calculated by the

following equations:

$$\mathbf{v}_g \sin \theta = \mathbf{v}_g + |\mathbf{v}_g \cdot \mathbf{n}| \cdot \mathbf{n} \quad (11)$$

$$\mathbf{v}_g \cos \theta = |\mathbf{v}_g \cdot \mathbf{n}| \cdot \mathbf{n} \quad (12)$$

2.6. Boundary conditions

The heat flux on the top surface of the molten pool includes laser heat flux, heat conduction loss, heat radiation loss and evaporation loss:

$$k \frac{\partial T}{\partial n} = q_l - h_c (T - T_{ref}) - \delta k_b (T^4 - T_{ref}^4) - 0.82 \frac{\Delta H^*}{\sqrt{2\pi M R T}} P_0 \exp \left(\Delta H^* \frac{T - T_v}{R T T_v} \right) \quad (13)$$

Where h_c and δ represent the convection coefficient and radiation emissivity, respectively. k_b and ΔH^* denote the Stefan-Boltzmann constant and vapor flow enthalpy, respectively.

The momentum boundary conditions in the normal and tangential directions of the free surface can be described as follows:

$$-\mathbf{P} + 2\mu \frac{\partial \mathbf{v}_n}{\partial n} = -P_r - P_{sta} + \frac{P_\sigma}{R_s} \quad (14)$$

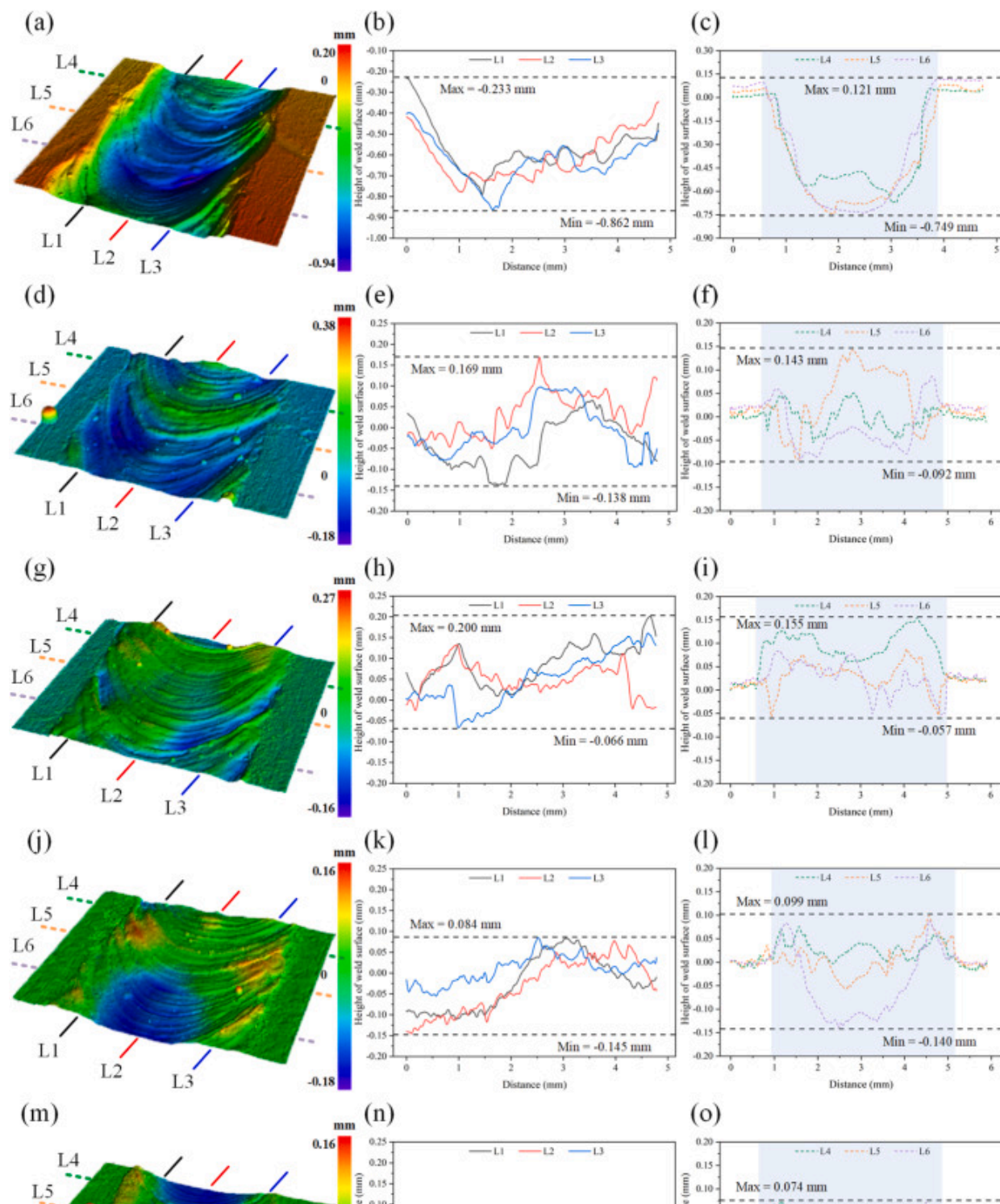
$$-\mu \frac{\partial \mathbf{v}_t}{\partial n} = \frac{\partial P_\sigma}{\partial T} \frac{\partial T}{\partial S} + P_{vap} \quad (15)$$

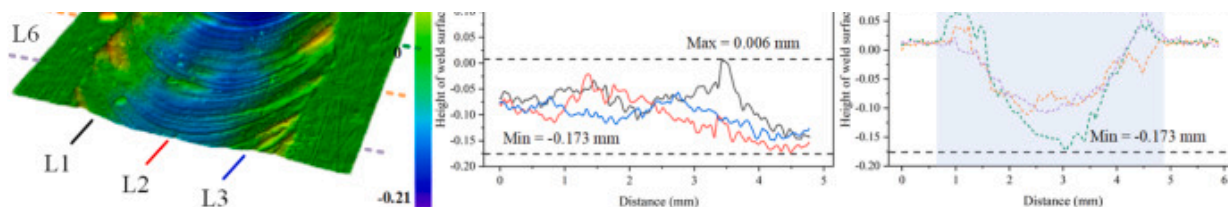
Where $\mathbf{v}_n$ and $\mathbf{v}_t$ are the normal melt velocity vector and the tangential melt velocity vector, respectively. R_s is the radius of the surface curvature, and S is the tangential melt velocity vector.

3. Results and discussion

3.1. Influence of ring power ratios on weld surface formation and internal porosity

The weld surface formation cloud diagram and surface heights statistics under various RPR are shown in Fig. 4. L1-L3 and L4-L6 are the surface height along and perpendicular to the weld direction. The black dashed lines indicate the maximum and minimum surface heights in different directions, while the blue-shaded region can be used as a reference to measure the weld width. As illustrated in the surface cloud map in Fig. 4(a), the RPR=0 (P_{C4}) exhibited a more visible weld surface collapse and an uneven surface. The addition of 500W of ring power ($P_{C3.5}P_{R0.5}$) produced the most weld spatters on the weld surface, but the surface collapse was improved, as displayed in Fig. 4(d). Surface spatter can be effectively reduced by further increasing the RPR, resulting in a more uniform surface texture and improved surface flatness, as demonstrated in Fig. 4(g)–(j), and (m). This was also supported by quantitative height statistics. The maximum surface height differences along the weld direction for various RPR are 0.629mm (RPR=0, P_{C4}), 0.307mm (RPR=12.5%, $P_{C3.5}P_{R0.5}$), 0.266mm (RPR=25%, $P_{C3}P_{R1}$), 0.229mm (RPR=37.5%, $P_{C2.5}P_{R1.5}$), and 0.179mm (RPR=50%, $P_{C2}P_{R2}$). Increasing the RPR significantly reduced the surface height difference. When RPR=50%, the maximum surface height difference was reduced by 71.5% compared with RPR=0, as seen in Fig. 4(b) and (n). The surface height perpendicular to the weld was employed to determine the degree of bulge and collapse of the weld surface. The maximum surface height differences for various RPR are 0.870mm (RPR=0), 0.235mm (RPR=12.5%), 0.212mm (RPR=25%), 0.239mm (RPR=37.5%), and 0.247mm (RPR=50%). When RPR=25% and RPR=50%, the maximum surface height difference was reduced by 75.6% and 71.6% respectively, compared to RPR=0. Therefore, in general, when RPR=50%, the weld surface exhibited the best flatness.



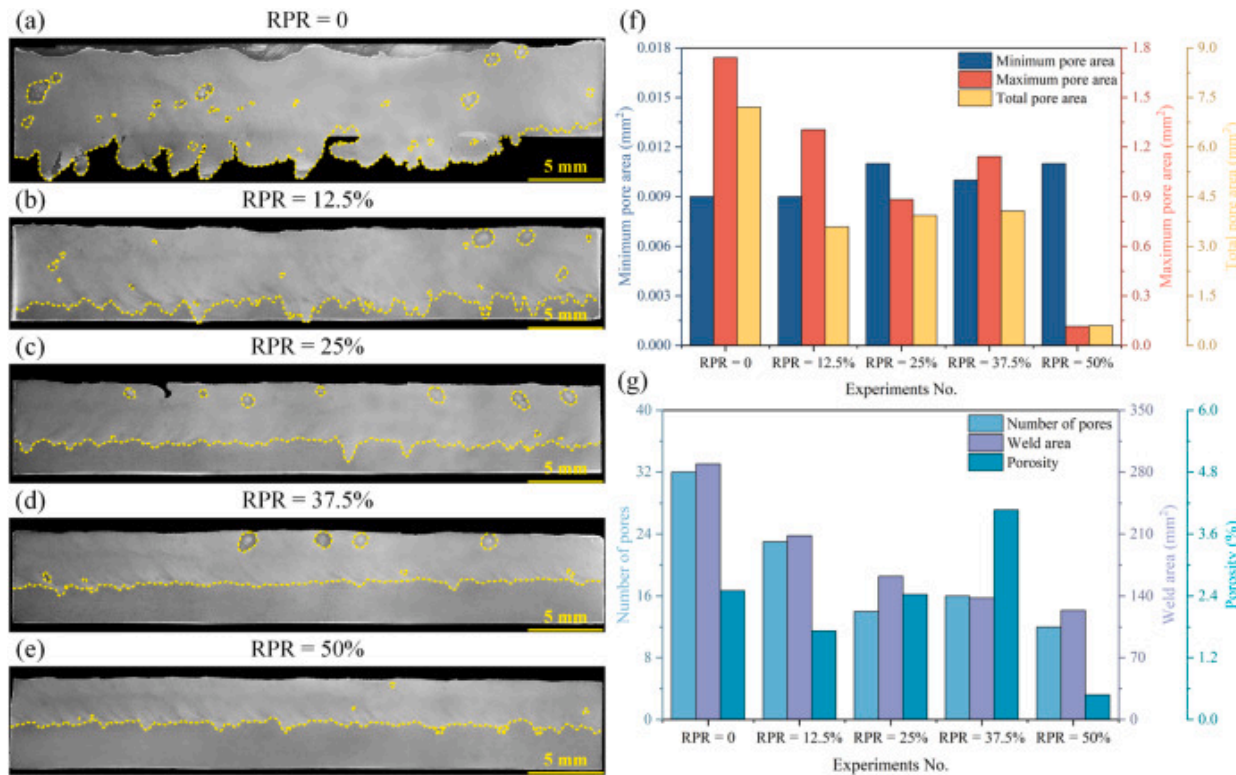


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Fig. 4. Weld formation cloud diagrams and surface height statistics along the weld direction and perpendicular to the weld direction under various ring power ratios (RPR). (a–c) RPR=0; (d–f) RPR=12.5%; (g–i) RPR=25%; (j–l) RPR=37.5%; (m–o) RPR=50%.

Fig. 5 shows the longitudinal profile of the weld and the porosity statistics under various RPR. The yellow dashed lines represent the pores and the fusion line at the bottom of the weld. When there was no ring power (RPR=0), the higher energy density of the core beam resulted in a full-penetration weld, accompanied by more severe surface collapse and a root hump. In addition, the weld had a considerable number of irregular pores, the majority of which were distributed at the middle and bottom of the weld, as indicated in Fig. 5(a). When the ring power was added and the proportion of the ring power increased, the fluctuation of the weld penetration line smoothed out, and the number of pores in the weld decreased dramatically. Fig. 5(b)–(e) displayed that the pore shape became more regular and concentrated in the top and middle of the weld, which was consistent with the experimental results of Li et al. [28]. The RPR=50% resulted in the fewest small pores in the weld. The size, number, and area ratio of the pores are demonstrated in Fig. 5(f) and (g). As the fraction of the ring power increased, the number and total area of pores varied similarly: initially decreasing significantly, then marginally increasing, and eventually decreasing to the minimum value when RPR=50%. The number of pores reduced by 62.5%, while the pore area decreased by 91.7%. When RPR=37.5%, the porosity grew inversely due to the significant reduction in the melting area of the additional ring beam. However, the RPR=50% still had the lowest porosity, which was only 0.48%, 80.8% lower than the 2.5% porosity of the RPR=0.



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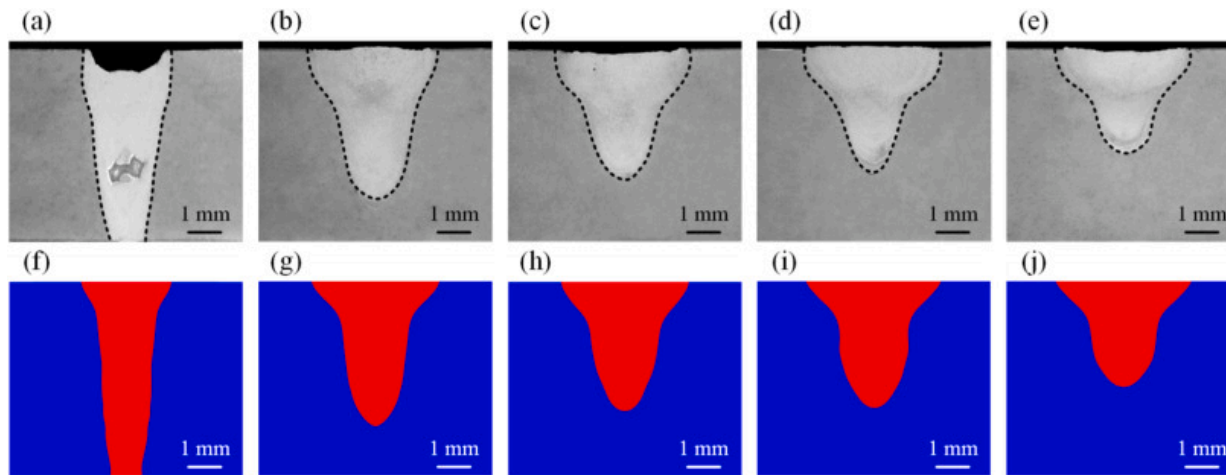
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Fig. 5. Weld longitudinal section and pore statistics under various ring power ratios (RPR). (a) Weld longitudinal section with RPR=0; (b) weld longitudinal section with RPR=12.5%; (c) weld longitudinal section with RPR=25%; (d) weld longitudinal section with RPR=37.5%; (e) weld longitudinal section with RPR=50%; (f) pore area under various RPR; (g) pore number, weld area and porosity under various RPR.

3.2. Influence of ring power ratios on keyhole dynamic behavior

Fig. 6(a–e) depicts the cross-sections of the welds obtained from the experiments for various RPR, whereas the corresponding simulation results are shown in Fig. 6(f–j). The weld area is highlighted with a black dashed line and a red region, respectively. To verify

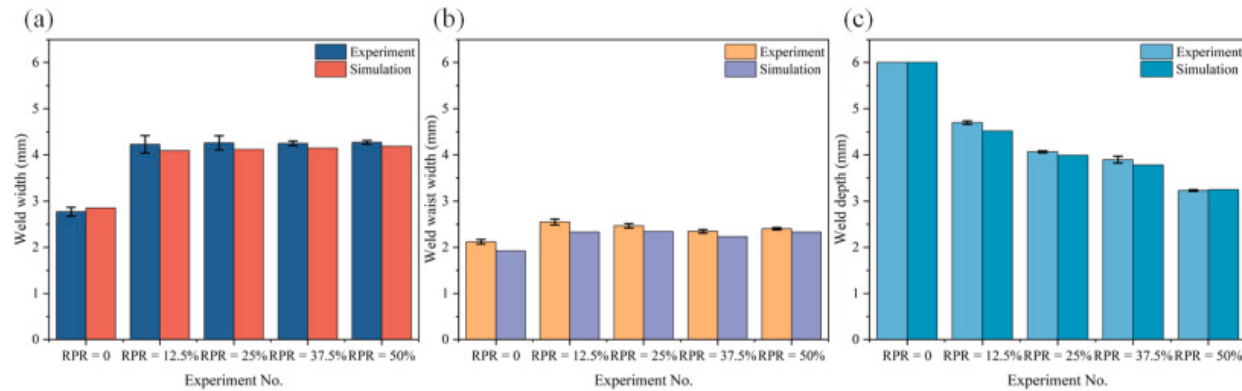
the reliability of the simulation calculation, Fig. 7 compares the cross-sectional dimensions obtained from the experiment and simulation. The error range for the weld width is 1.87%–3.31%, with a maximum difference of 0.14 mm (RPR=25%, experimental value: 4.26 mm, simulated value: 4.12 mm). The error range for weld waist is 2.92%–9.43%, with a maximum difference of 0.22 mm (RPR=12.5%, experimental value: 2.55 mm, simulated value: 2.33 mm). The error range for weld penetration is 0–3.83%, with a difference of 0.18 mm (RPR=12.5%, experimental value 4.7 mm, simulated value 4.52 mm). This indicates good agreement between the experimental and simulated results, implying that this model can be used to further explore keyhole dynamics and molten pool heat-flux distribution under various operating conditions.



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Fig. 6. Weld cross-section findings with various ring power ratios (RPR) produced through experiment and simulation. (a) and (f) experimental and simulated cross-section with RPR=0; (b) and (g) experimental and simulated cross-section with RPR=12.5%; (c) and (h) experimental and simulated cross-section with RPR=25%; (d) and (i) experimental and simulated cross-section with RPR=37.5%; (e) and (j) experimental and simulated cross-section with RPR=50%.

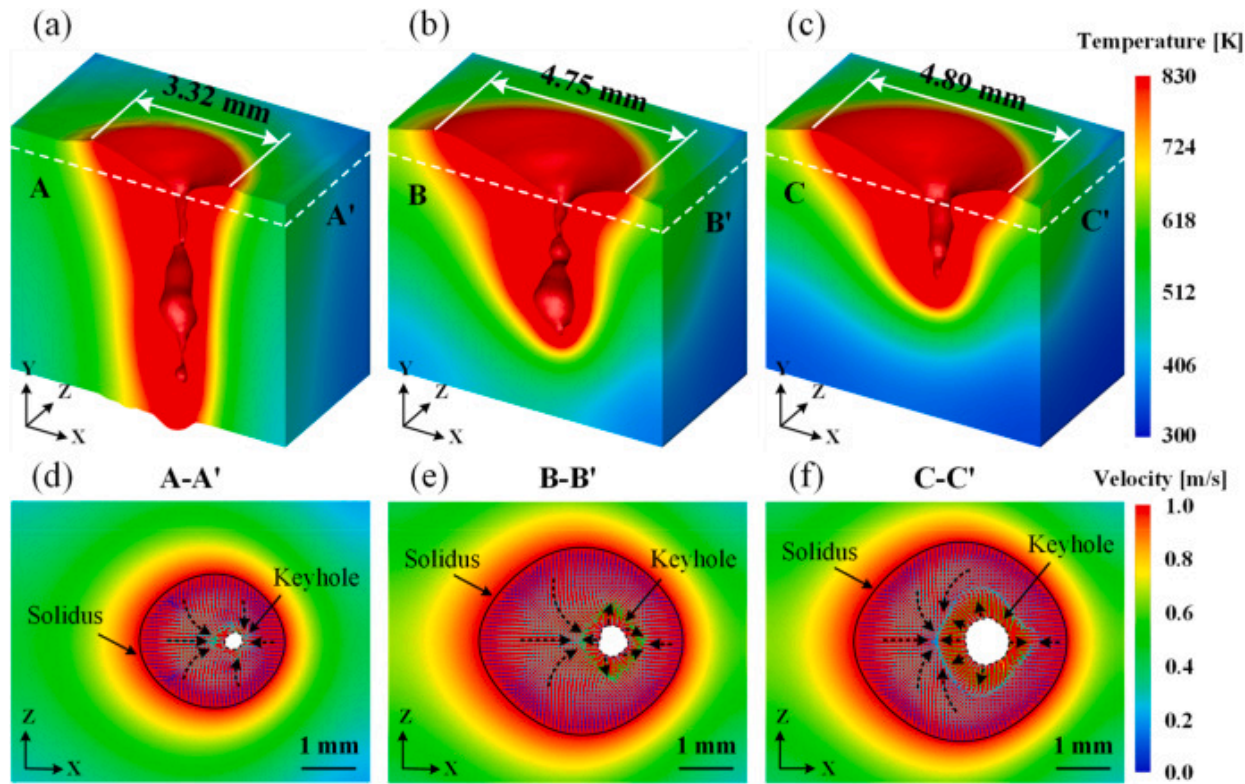


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Fig. 7. Comparison of experimental and simulated weld dimensions under various ring power ratios (RPR). (a) Weld width; (b) weld waist width; (c) weld depth.

Fig. 8 exhibits the three-dimensional temperature distribution of the molten pool in steady state, as well as the heat-flux distribution at the horizontal cross-section of the keyhole entrance for various RPR. The white dashed line represents the cross section of the keyhole entrance, and the black solid line demarcates the molten pool boundary. Since the core power and ring power focus on contributing to the depth of penetration and the length and width of the molten pool, respectively [41], therefore, under the condition of constant total input power, it can be found that the RPR=50% had the longest molten pool (4.89mm) and the smallest depth of penetration. In contrast, the molten pool length with RPR=0 was the shortest (only 3.32mm), but the melting depth was the largest, as illustrated in Fig. 8(a–c). When RPR=0, the horizontal cross-section at the keyhole entrance only flowed exclusively from the border of the molten pool toward the keyhole, as seen in Fig. 8(d). Adding a ring beam created reverse flow at the keyhole entrance, and a higher RPR broadened the range of reverse flow. This effectively resists the influence of melt on the keyhole entrance, minimized keyhole closure, and resulted in a wider and more stable keyhole channel, as shown in Fig. 8(e) and (f).



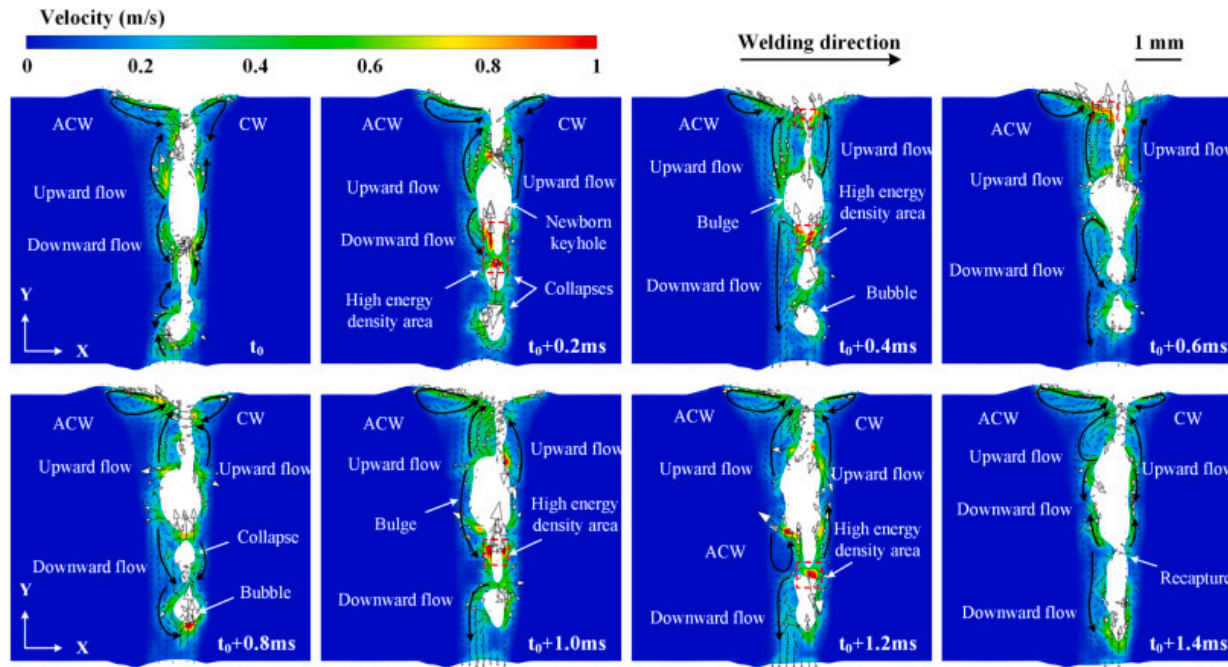
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Fig. 8. (a–c) Three-dimensional temperature distribution of the molten pool with ring power ratio (RPR) of 0, 25% and 50%, respectively; (d–f) heat-flux distribution in the horizontal section at the keyhole entrance with RPR of 0, 15% and 50%, respectively.

The dynamic keyhole and flow velocity distribution at different moments of the central section ($Z=3.6\text{ mm}$) when $\text{RPR}=0$ are displayed in Fig. 9. The high energy density input and the slender keyhole channel allowed the keyhole extremely prone to frequent collapse and reconnection, and the keyhole and flow velocity distribution exhibited highly transient characteristics [42]. At the initial, relatively regular keyhole moment (t_0), multiple flows occurred in the depth direction. Marangoni flows dominated the near-surface molten pool, with several upward and downward flows in the middle and bottom

of the keyhole, respectively. The downward flow in the middle and the upward flow at the bottom had a combined influence on the constriction of the keyhole, generating multiple collapses, bulges, and newborn keyholes, as seen at $t_0+0.2\text{ms}$ and $t_0+0.4\text{ms}$. The newborn keyhole focused the laser beam at the bottom to produce a high-energy-density area. The intense evaporation pushed the protrusion upward while generating a downward flow. The evaporation of the liquid metal layer caused partial collapse, which was recaptured by the keyhole to achieve further drilling. The uncaptured collapsed portion will form a bubble unless the keyhole can recapture it before the solidification front, such as at $t_0+1.2\text{ms}$ and $t_0+1.4\text{ms}$. It can be observed that during the entire cycle, an upward flow occurred along the keyhole wall in the middle keyhole, which impacted the keyhole entrance together with the anticlockwise (ACW) and clockwise (CW) near the surface, forcing it to close. In addition, the strong metal vapor shear eventually led the molten pool surface to expand and contract frequently, which was the primary source of spatter and uneven surface, as demonstrated in [Fig. 4\(a–c\)](#). Furthermore, when bubbles emerged owing to a fracture at the bottom keyhole, they were suppressed by the downward flow and cannot float up, which enhanced the probability of them being trapped by the solidification front and eventually forming pore defects. As a result, pore defects mainly were concentrated at the middle and bottom, as displayed in [Fig. 5\(a\)](#).



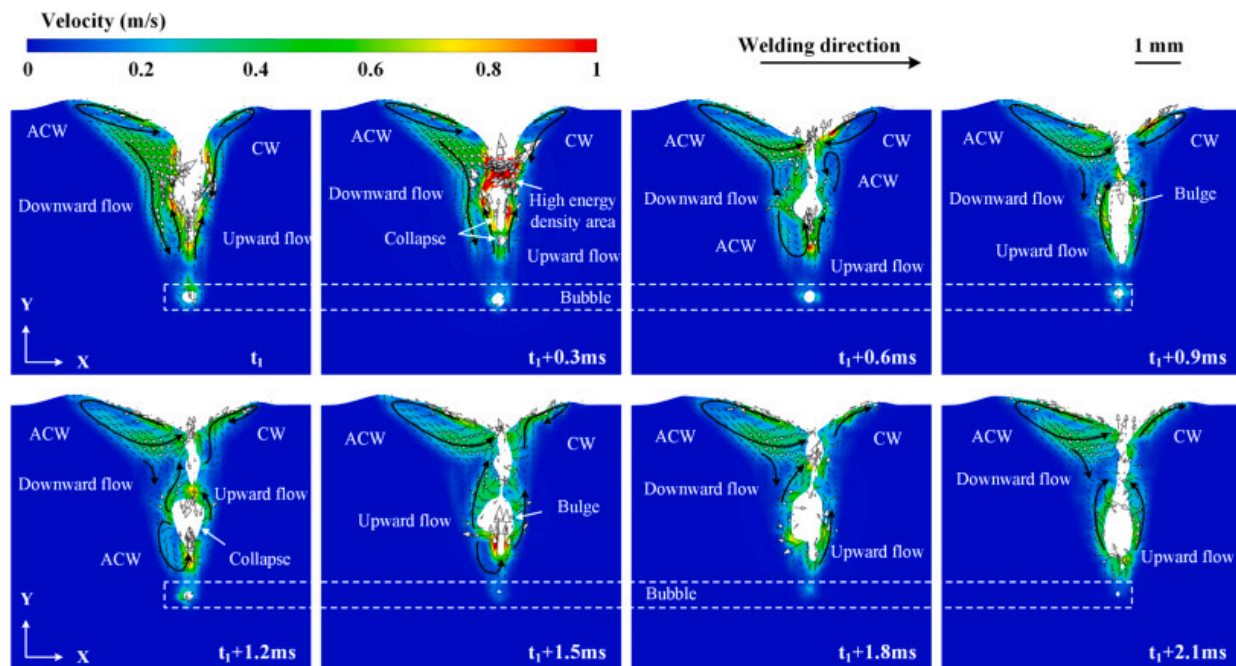
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Fig. 9. Velocity distribution in the central section ($Z=3.6\text{mm}$) with ring power ratio (RPR) of 0.

Fig. 10, Fig. 11 describe the dynamic keyhole and flow velocity distribution at various moments in the central section ($Z=3.6\text{mm}$) for RPR=25% and RPR=50%, respectively. Compared with the multiple irregular flows of RPR=0, the RPR=25% and RPR=50% exhibited long-distance regular flow, with RPR=50% having the fewest tributaries in the depth direction. When the keyhole was somewhat regular, the downward flow from the top surface to the bottom was dominant, while there was also an upward flow along the keyhole wall from the bottom surface, such as t_1 ms and $t_2+4.2$ ms. The bidirectional flow along the keyhole wall converged and struck the keyhole wall, causing the keyhole to shrink and fracture, followed by collapse and the formation of a new keyhole, such as $t_1+0.3$ ms and $t_2+0.6$ ms. This process was similar to RPR=0. In contrast, because of the

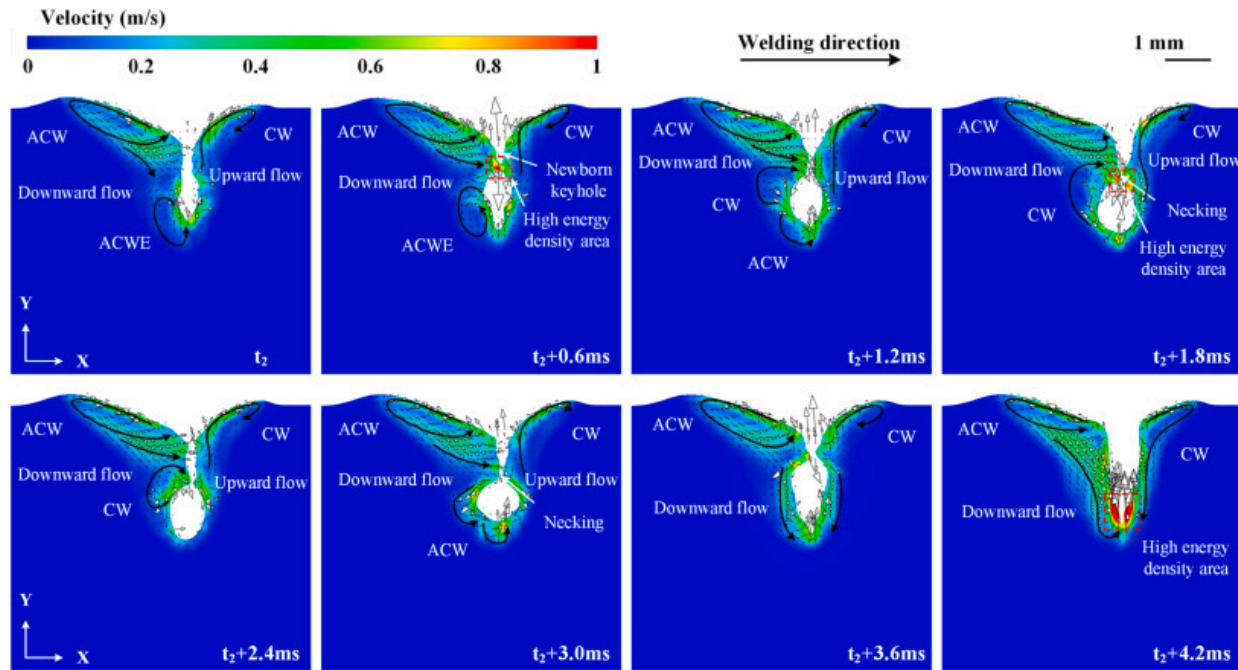
larger keyhole entrance and smooth keyhole wall, the metal vapor generated by the high energy density can escape quickly. The resulting upward flow mixed with the ACW along the wall, and the downward flow produced a local anticlockwise eddy (ACWE) at the bottom, along the collapsed keyhole wall, such as $t_1+0.6\text{ms}$ and $t_2+0.6\text{ms}$. When the high-energy-density evaporated liquid metal layer and recaptured the collapsed keyhole, the laser energy reached the keyhole bottom, where the resulting upward flow from the bottom intersected with the downward flow and hit the keyhole protrusion, repeating the collapse and reconnection. During this process, the bubbles generated at the initial moment of $\text{RPR}=25\%$ were repeatedly polished by the upward and downward flows, becoming more regular in shape and smaller in size due to the partial recapture of the bubbles. The bubbles were frozen into tiny pore defects by the local low temperature created by the absence of laser irradiation at the bottom [43]. It was discovered that the flow's effect on the keyhole was more pronounced with $\text{RPR}=50\%$ and was closer to the middle and lower part of the keyhole throughout the entire cycle. While upward flow from the middle keyhole still affected the keyhole entrance, along with the ACW and CW, with $\text{RPR}=25\%$, the timing of this influence was significantly reduced compared to $\text{RPR}=0$. This reduction was most pronounced with $\text{RPR}=50\%$. This significantly reduced the effect of the upward flow from the middle on the molten pool surface. Combined with the weakening of vapor shearing by the wide keyhole entrance, this effectively mitigated the expansion and contraction of the molten pool, resulting in a smoother and more uniform surface, as shown in Fig. 4(g-i) and (m-o). Thanks to the increased upward flow from the bottom of the molten pool, with $\text{RPR}=25\%$ and $\text{RPR}=50\%$, bubbles rose faster, increasing their probability of escape. Even if they were ultimately captured by the gas-liquid interface, porosity defects were concentrated near the molten pool surface, as shown in Fig. 5(c). However, due to the lower penetration depth of $\text{RPR}=50\%$, the probability of bubble escape was maximized, and porosity defects were minimized, as illustrated in Fig. 5(e).



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Fig. 10. Velocity distribution in the central section ($Z=3.6\text{mm}$) with ring power ratio (RPP) of 25%.



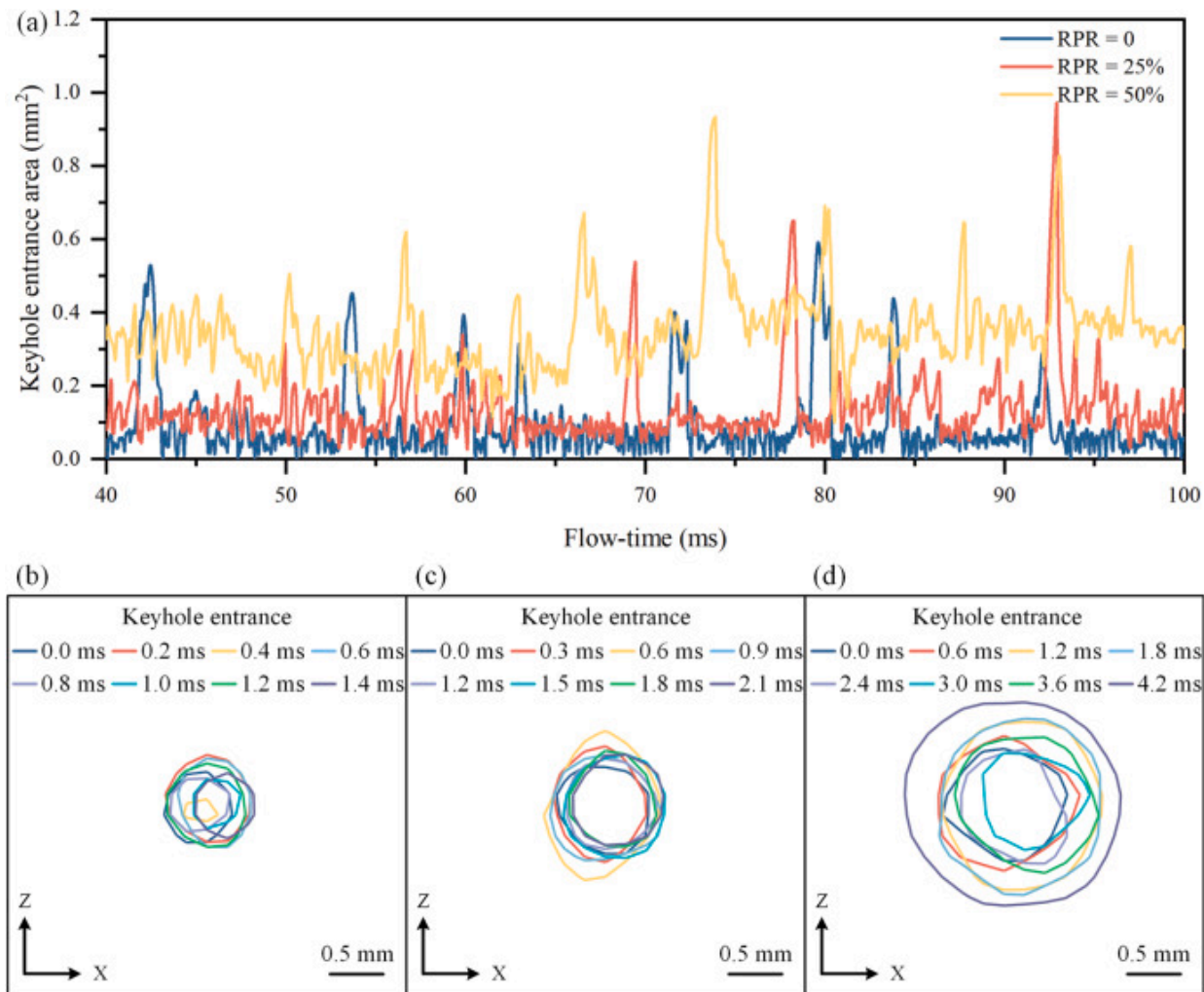
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Fig. 11. Velocity distribution in the central section ($Z=3.6\text{mm}$) with ring power ratio (RPR) of 50%.

Fig. 12 extracts the keyhole entrance area and entrance profile at different moments for various RPR. The average keyhole entrance area for various RPR were 0.09mm^2 (RPR=0), 0.14mm^2 (RPR=25%), and 0.37mm^2 (RPR=50%), indicating that the keyhole entrance area expanded dramatically with the increase in RPR, as shown in Fig. 12(a). A wider keyhole entrance effectively alleviated vapor shearing on the molten pool, preventing uneven forming caused by the expansion and contraction of the molten pool. Fig. 12(b) and (d) show the horizontal cross-section keyhole entrance contours extracted from Fig. 9, Fig. 10, Fig. 11 at the corresponding moments. From another perspective, a significant increase in the keyhole entrance area was observed with RPR=25% and RPR=50%. Additionally, as the RPR increased, the keyhole entrance contour became more regular and more concentric,

effectively avoiding keyhole swing along the welding direction and providing greater keyhole stability.



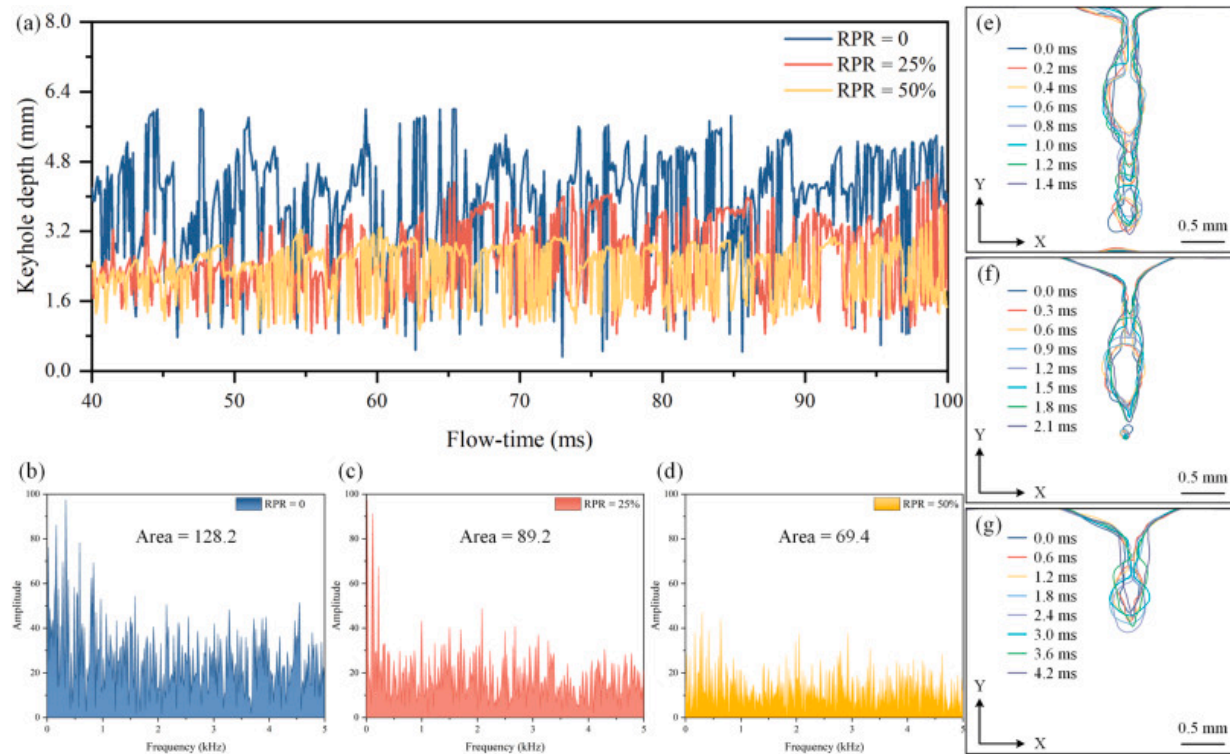
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Fig. 12. Changes in keyhole entrance with various ring power ratios (RPR) at different moments. (a) Keyhole entrance area fluctuation curve from 40ms to 100ms; (b–d)

keyhole entrance contours at different moments with RPR=0, RPR=25%, and RPR=50%, respectively.

The variations in keyhole depth and wall profile at different moments for various RPR are depicted in Fig. 13. The comparatively constant range of 40–100ms was used for the statistical analysis of the keyhole depth. The standard deviations of the keyhole depth for various RPR were 1.22mm (RPR=0), 0.86mm (RPR=25%), and 0.65mm (RPR=50%), indicating that the fluctuation range of the keyhole depth decreased with increasing RPR. The reduction effect was most significant with RPR=50%, where the standard deviation was reduced by 46.72% compared to RPR=0. Fig. 13(b–d) displays the Fourier transforms of the keyhole depth curves for three various RPR. The peak strength and the curve envelope area can be used to assess stability. The envelope areas of the Fourier transform curves for various RPR were 128.2 (RPR=0), 89.2 (RPR=25%), and 69.4 (RPR=50%), respectively. It was evident that both the peak strength and envelope area decreased significantly with the increase in RPR, with RPR=50% exhibiting the best keyhole stability. The keyhole wall contours at the corresponding moments in Fig. 9, Fig. 10, Fig. 11 are shown in Fig. 13(e–g). There were fewer protrusions and collapses in the keyhole wall with RPR=50%. The bottom contour curve of the complete keyhole has the maximum repetition rate.



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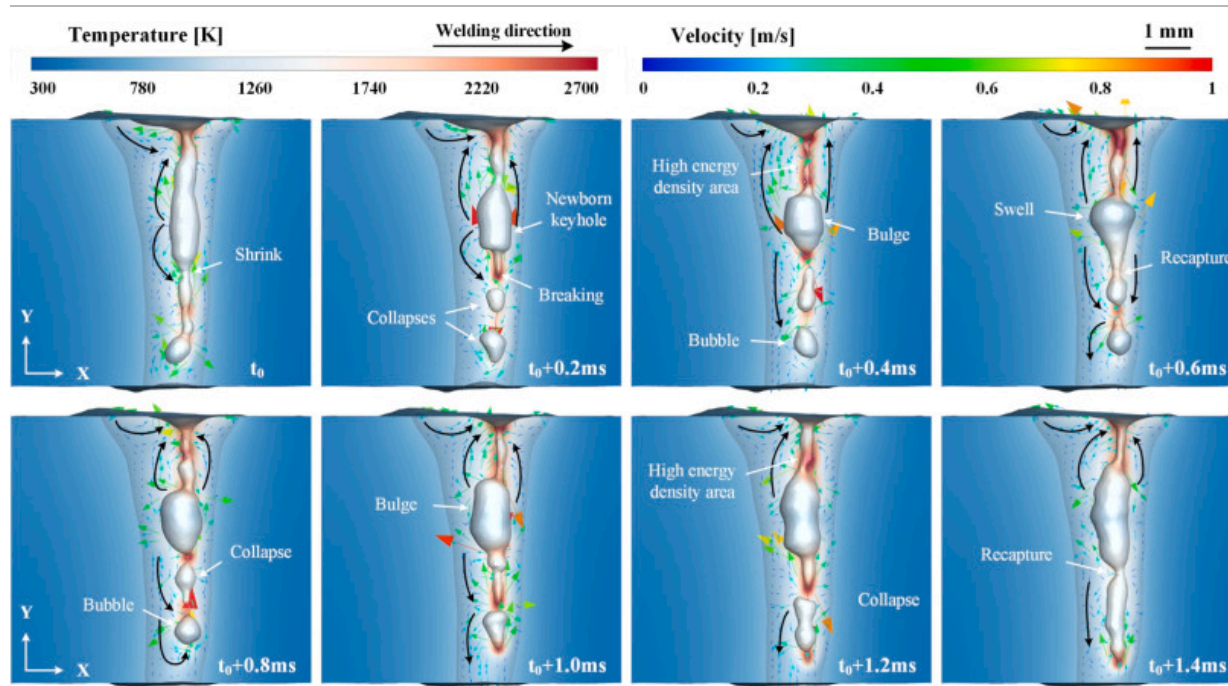
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Fig. 13. Changes in keyhole depth with various ring power ratios (RPR) at different moments. (a) Keyhole depth fluctuation curve from 40ms to 100ms; (b–d) Fourier transforms of keyhole depth with RPR=0, RPR=25%, and RPR=50%, respectively; (e–g) keyhole wall profile at different moments with RPR=0, RPR=25%, and RPR=50%, respectively.

3.3. Influence of ring power ratios on thermal-fluid characteristics of molten pool

Fig. 14 displays the heat-flux distribution of the keyhole and the center cross-section of the molten pool ($Z=3.6\text{mm}$) for RPR=0. The molten pool is delineated by the gray

translucent region. At the initial, relatively regular keyhole moment (t_0), the laser energy reached the bottom of the keyhole, where a localized high-temperature region existed. This affected the constriction of the keyhole and generated an upward flow along the keyhole wall, which led to fracture and multi-segment collapse ($t_0+0.2\text{ms}$). Strong upward and downward flows were produced along the keyhole wall ($t_0+0.4\text{ms}$) as a result of the laser energy being preferentially absorbed by the bottom of the newborn keyhole and new protrusions, preventing it from reaching deeper into the keyhole. The evaporation of the liquid metal layer allowed partially collapsed layer to be recaptured, allowing the laser energy to reach the middle and bottom parts ($t_0+0.6\text{ms}$). However, due to the high energy density, the elongated keyhole channel and the irregular protrusions shielding the energy, causing these processes to recur repeatedly. During this procedure, bubbles were momentarily formed in the partially collapsed keyhole that was not retrieved by the keyhole. If these bubbles were not recovered before the arrival of the solidification front, they would form porosity defects. As can be observed, due to the lack of laser energy at the keyhole bottom, the upward flow was insufficient, and there was no adequate support for the bubbles to float upward, which increased the probability of bubbles forming porosity defects at the bottom. Furthermore, the laser energy was mainly distributed in the upper and middle parts of the keyhole. The stronger upward flow also posed more severe challenges to the keyhole entrance and the stability of the molten pool surface.



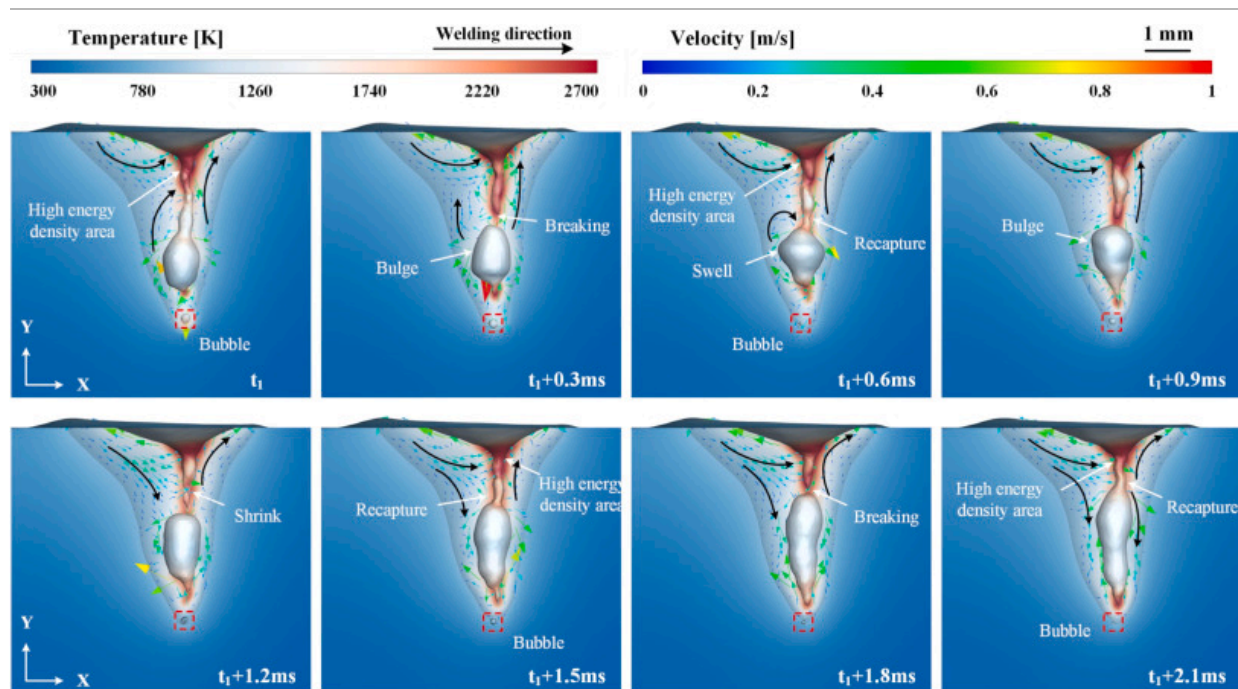
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Fig. 14. Heat-flux distribution in the keyhole and the central section of the molten pool ($Z=3.6\text{mm}$) with ring power ratio (RPR) of 0.

Compared with $\text{RPR}=0$, the repeated collapse caused by keyhole oscillation was alleviated with $\text{RPR}=25\%$ and $\text{RPR}=50\%$, as illustrated in Fig. 15, Fig. 16. The ring beam provided a wider molten pool range, which overall exhibited a regular flow of fewer, longer streams, effectively avoiding collision on the keyhole wall and preserving keyhole stability. Moreover, a wider keyhole entrance and channel were provided with $\text{RPR}=25\%$ and $\text{RPR}=50\%$, enabling laser energy to radiate to the middle and bottom of the keyhole, resulting in an upward flow along the keyhole wall at the bottom. Additionally, the bottom protrusion was entrained to move upward by the powerful metal vapor, which further widened the central keyhole channel. As can be seen, compared to the influence of the flow on the top of the keyhole with $\text{RPR}=0$, a similar effect was only apparent at t_1

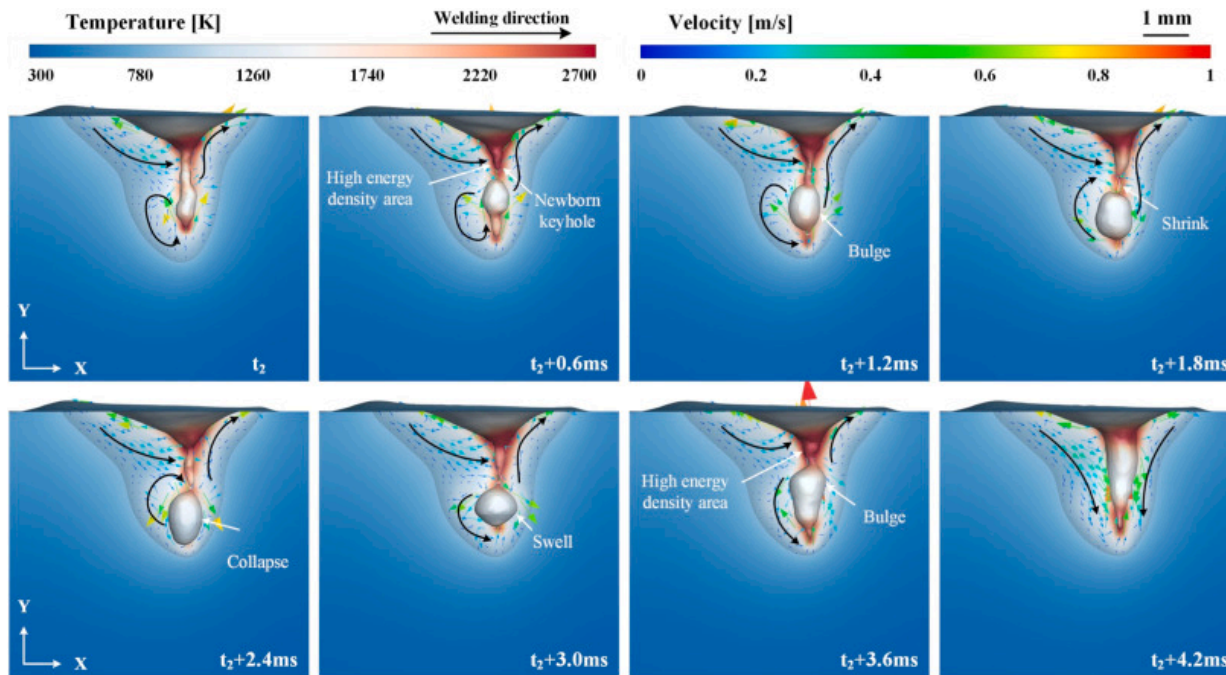
and $t_1+0.3\text{ms}$ in Fig. 15 with RPR=25%, significantly reducing the impact duration. Meanwhile, the effect of the upward flow along the bottom bulge on the keyhole middle is only observed at $t_2+1.8\text{ms}$ and $t_2+2.4\text{ms}$ in Fig. 16 with RPR=50%. These factors significantly contribute to maintaining keyhole stability, thereby enhancing the amount of laser energy absorbed by the bottom of the keyhole. The bubble appeared at the initial moment in Fig. 15 was continuously polished to a more regular shape before being captured, achieving its lowest surface energy [44]. Because the protrusion above the bubble shielded the laser energy from reaching the bubble's bottom, the temperature near the bubble was lower, establishing a low-temperature zone. As the keyhole and the bottom of the molten pool fluctuated, the bubble was partially captured by the liquid-solid interface ($t_1+1.5\text{ms}$). The uncaptured portion, unable to fill the liquid phase, ultimately developed a porosity defect ($t_1+2.1\text{ms}$). No porosity defects resulting from keyhole oscillation were observed with RPR=50%, and the collapsed keyholes were recaptured during the subsequent further drilling process. The wider molten pool at the bottom gave the bubbles enough time to float up and escape from the molten pool.



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Fig. 15. Heat-flux distribution in the keyhole and the central section of the molten pool ($Z=3.6 \text{ mm}$) with ring power ratio (RPR) of 25%.

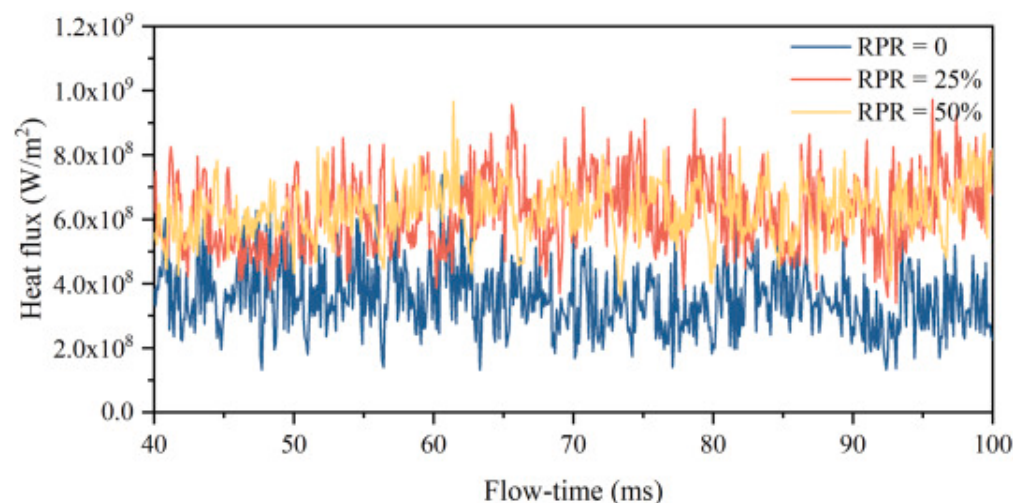


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Fig. 16. Heat-flux distribution in the keyhole and the central section of the molten pool ($Z=3.6\text{mm}$) with ring power ratio (RPR) of 50%.

Fig. 17 displays the average heat flux variation at the keyhole wall from 40 to 100ms. The highest average heat flux ($6.25 \times 10^8 \text{W/m}^2$) and the smallest standard deviation ($8.76 \times 10^7 \text{W/m}^2$) were found with RPR=50%. The average heat flux of $3.55 \times 10^8 \text{W/m}^2$, with a standard deviation of $1.08 \times 10^8 \text{W/m}^2$ for RPR=0. Smaller average heat flux fluctuations represented a more stable keyhole [45]. Furthermore, a larger keyhole entrance attributed to an increase in RPR permitted more laser energy to enter the keyhole interior and undergo multiple reflections, resulting in greater energy absorption by the wall.



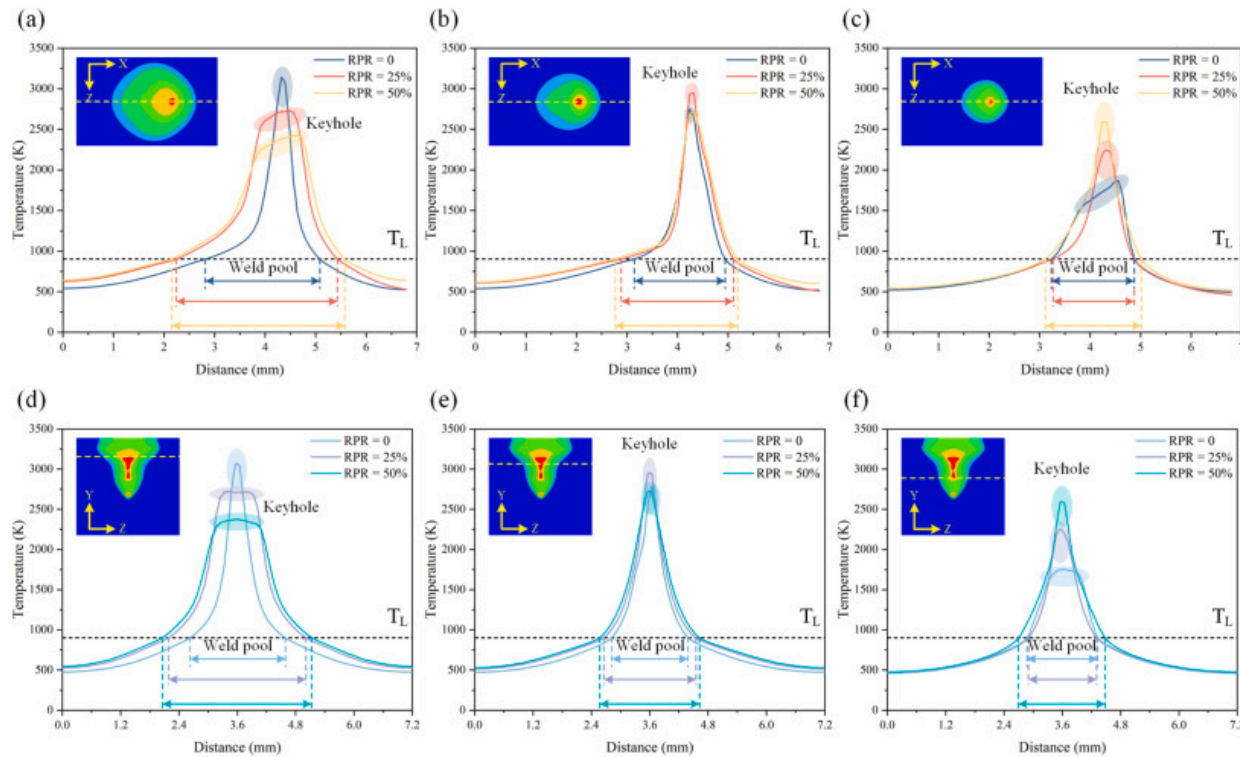
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Fig. 17. Changes in average heat flux at different moments under various ring power ratios (RPR).

The temperature distribution for various RPR was further examined. Temperature curves were extracted along the penetration direction at three different locations, along the horizontal weld center and along the cross-sectional weld center, as seen in Fig. 18. The peak of the curve represented the keyhole size at the corresponding location. It was evident that the temperature gradient was steep near the keyhole and gradually flattened as it moved away from it. Near the top of the keyhole ($Y=0.5\text{mm}$), the RPR=0 had the highest peak temperature (over 3000K) and the steepest temperature gradient from the keyhole to the border of the molten pool. As the RPR increases, the peak temperature and the temperature gradient from the center to the border of the molten pool diminished significantly, reaching a minimum with RPR=50%, as demonstrated in Fig. 18(a) and (d). Due to the unstable keyhole and multiple protrusions of RPR=0, the peak temperature of RPR=0 dramatically dropped with increased sampling depth, as shown in Fig. 18(b) and (c). This further demonstrated that the upper part of the keyhole was where the majority

of the laser energy, corresponding to $RPR=0$, was mainly distributed. As indicated in Fig. 18(e) and (f), the peak temperatures of $RPR=25\%$ and $RPR=50\%$ both rose first and then fell. The peak temperature differences at three locations with various RPR were 1274K ($RPR=0$), 707K ($RPR=25\%$), and 299K ($RPR=50\%$), respectively. It could be found that the peak temperature differences with $RPR=25\%$ and $RPR=50\%$ were significantly reduced, indicating that the laser energy can reach the middle and bottom of the keyhole wall, and the energy distribution uniformity in the depth direction of the keyhole was greatly improved.



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Fig. 18. Temperature distribution at different locations with various ring power ratios (RPR). (a–c) Temperature distribution along the horizontal weld center at $Y=0.5\text{ mm}$,

Y=1.0mm, and Y=2.0mm, respectively; (d-e) temperature distribution along the cross-section weld center at Y=0.5mm, Y=1.0mm, and Y=2.0mm, respectively.

3.4. Influence of ring power ratios on solidification characteristics

Considering the influence of the rapid solidification process in laser welding on the final microstructure, the solidification characteristics of various RPR were further investigated based on the temperature distribution at different locations in the molten pool.

Throughout the solidification process, the temperature gradient G , the solidification rate R , and their combination determine the morphology and size of the final microstructure [46]. G and R can be calculated using the following equations [47]:

$$G = \sqrt{G_x^2 + G_y^2 + G_z^2} \quad (15a)$$

$$R = V |G_x| / G \quad (16)$$

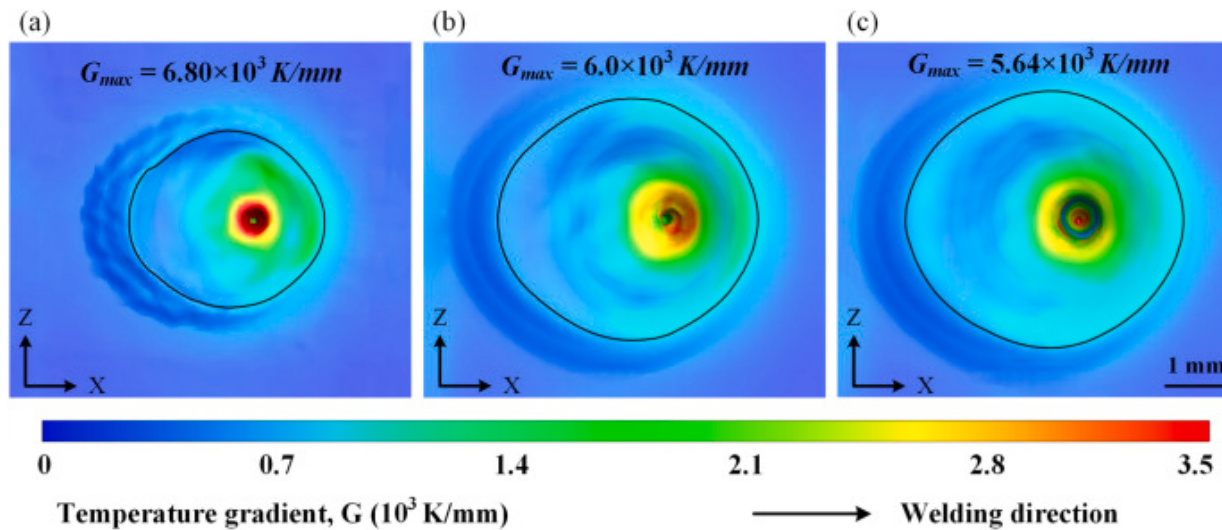
Where G_x , G_y and G_z are the temperature gradients in the x, y and z directions, respectively. V is the laser scanning speed.

The cooling rate, GR , determines the grain size of the solidified structure, while the morphological parameter, G/R , characterizes the degree of undercooling and establishes the morphology of the solidified structure. The G/R ratio can be used to assess the stability of the solid-liquid interface, determining whether it forms a stable planar structure or develops into a cellular or dendritic structure. The instability criterion for the solidification front can be expressed as follows [48]:

$$G/R < \Delta T_E / D_L \quad (17)$$

ΔT_E is the temperature difference between the solidus and liquidus, which in this study is 73K. D_L is the diffusion coefficient of magnesium in liquid aluminum, which is approximately $3 \times 10^{-3} \text{ mm}^2/\text{s}$ [49]. Therefore, for this study, the threshold value of G/R is approximately $2.43 \times 10^4 \text{ ks/mm}^2$.

The distribution of the temperature gradient G at the top of the molten pool for various RPR was displayed in Fig. 19. The solid black line marks the molten pool boundary. For RPR=0, due to the narrow keyhole channel, the high energy density produced the most significant temperature gradient, reaching a maximum of $6.80 \times 10^3 \text{ K/mm}$. The keyhole channel was widened by increasing RPR, which permitted the laser energy to enter the keyhole more deeply and produce a more even distribution of energy. The maximum temperature gradient decreased to $6.0 \times 10^3 \text{ K/mm}$ and $5.64 \times 10^3 \text{ K/mm}$ for RPR=25% and RPR=50%, respectively.



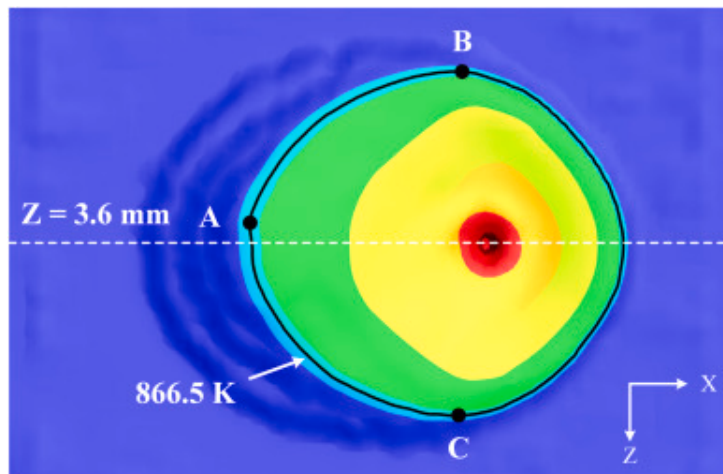
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Fig. 19. Temperature gradient distribution on the top surface of the molten pool with various ring power ratios (RPR). (a) RPR=0; (b) RPR=25%; (c) RPR=50%.

The morphology and size of the solidification structure at different locations may be predicted using G and R in the solidification zone. In this study, the isotherm $T=866.5 \text{ K}$ ($(T_S+T_L)/2$) was chosen as the measurement location for G and R . Data were extracted at intervals of unit grid size along the Z direction until the maximum Z -direction position, as shown in Fig. 20. Fig. 21 illustrates the changes in horizontal solidification characteristic

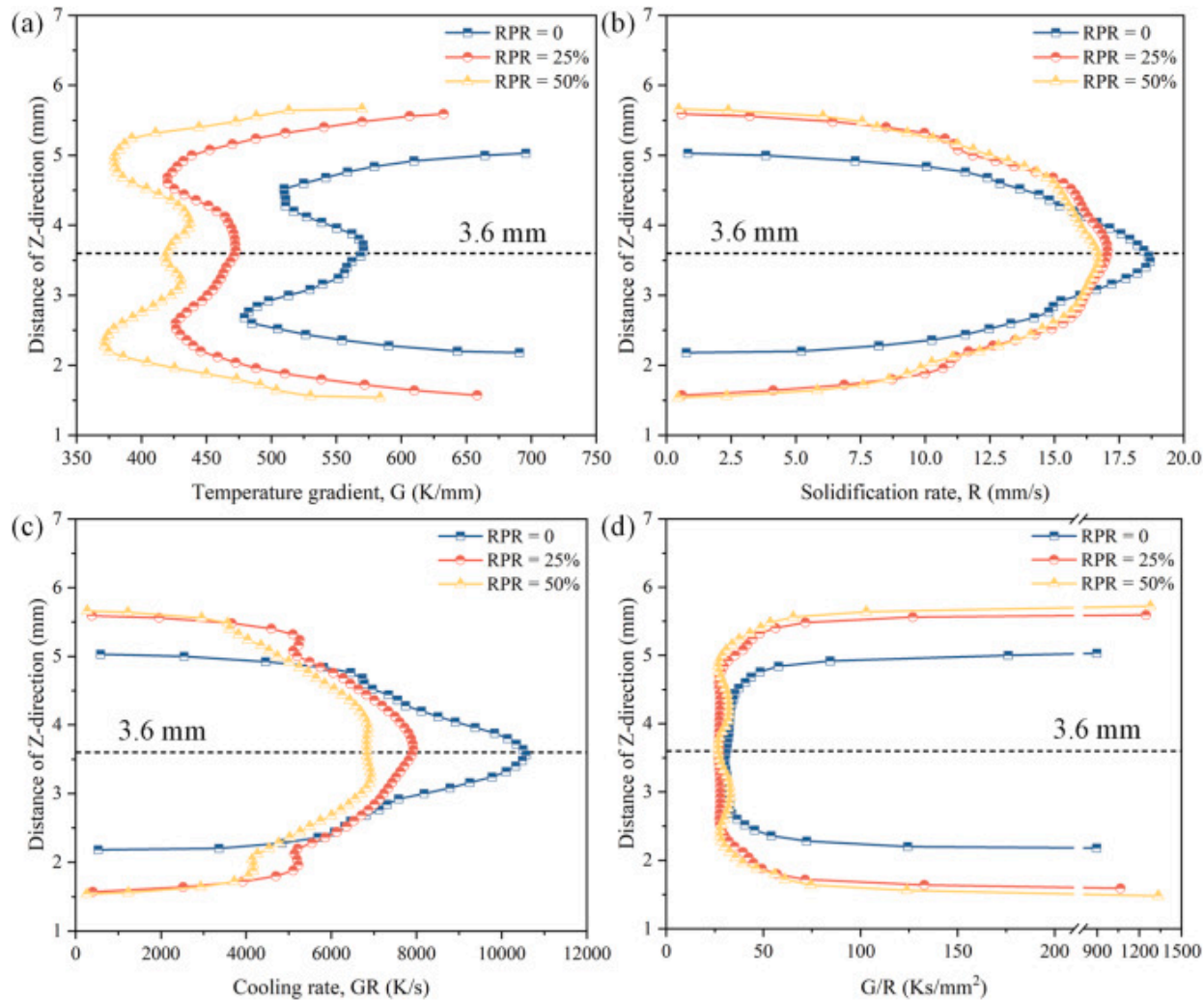
parameters for various RPR, with $Z=3.6\text{mm}$ representing the weld centerline. The RPR=25% and RPR=50% significantly reduced G , with the reduction effect being more pronounced at higher RPR. The temperature gradient at the fusion boundary was substantially greater than that at the weld center, as shown in Fig. 21(a). The normal vector of the liquid-solid interface was parallel to the welding direction at the center of the weld and perpendicular to it at the fusion boundary. Therefore, the solidification rate R was the largest at the center and close to 0 at the fusion boundary. As seen in Fig. 21(b), RPR=25% and RPR=50% slightly reduced the R at the weld center and fusion boundary, and the R values at these two locations were similar. A similar trend was exhibited in the cooling rate GR , which progressively rose from the fusion boundary to the weld center. As indicated in Fig. 21(c), the GR with RPR=0 was substantially higher than that with RPR=25% and RPR=50% at the weld center, although the three were comparable at the fusion boundary. The undercooling G/R value dropped dramatically from the fusion boundary to the weld center. The RPR=50% had the highest G/R value, approximately 1350ks/mm^2 , which was far below the G/R threshold ($2.43 \times 10^4\text{ks/mm}^2$). Therefore, only cellular and dendritic microstructures were expected to exist in the solidified weld.



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Fig. 20. Schematic diagram of the sampling method for temperature gradient G and solidification rate R (the solid black line represents the isotherm of $(T_S + T_L)/2$).



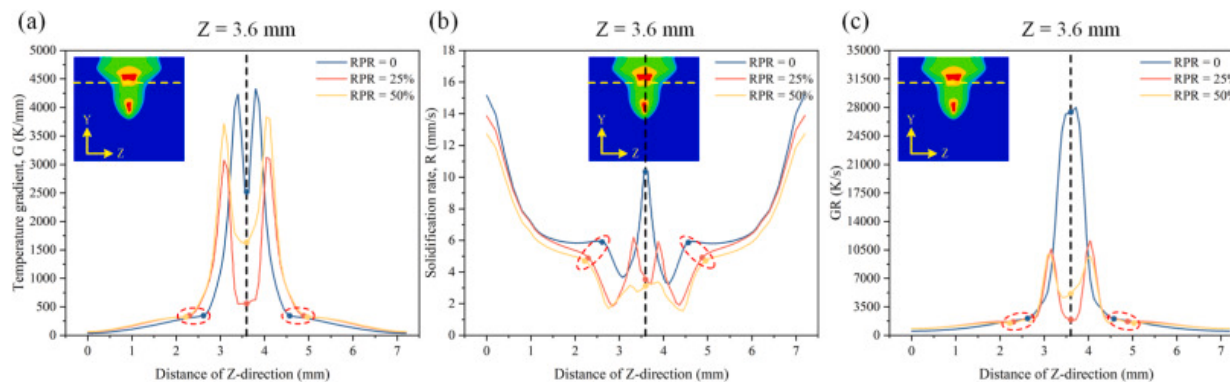
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Fig. 21. Variation of solidification parameters along the isotherm of 866.5K under various ring power ratios (RPR). (a) Temperature gradient G ; (b) solidification rate R ; (c) cooling

rate GR; (d) undercooling G/R .

To investigate the solidification characteristics at the center and fusion boundary of the weld cross-section, further analysis of the solidification parameters along the Z-direction at $Y=0.8\text{ mm}$ in the cross-section was carried out. As displayed in Fig. 22, at the center of the cross-section, the $RPR=0$ exhibited the highest G and R values. The G and R values decreased for both $RPR=25\%$ and $RPR=50\%$, with $RPR=50\%$ having a higher G value and a slightly lower R value than $RPR=25\%$. $RPR=25\%$ had the lowest GR value. At the fusion boundary, the G values were similar across various RPR , with $RPR=0$ having the highest R value and $RPR=50\%$ having the lowest GR value. Since the GR value dictated the grain size of the solidified structure (faster cooling rate led to smaller grain size) [50], it was anticipated that $RPR=0$ would result in the smallest grain size at the center of the cross-section. At the same time, $RPR=50\%$ would produce the largest grain size at the fusion boundary.

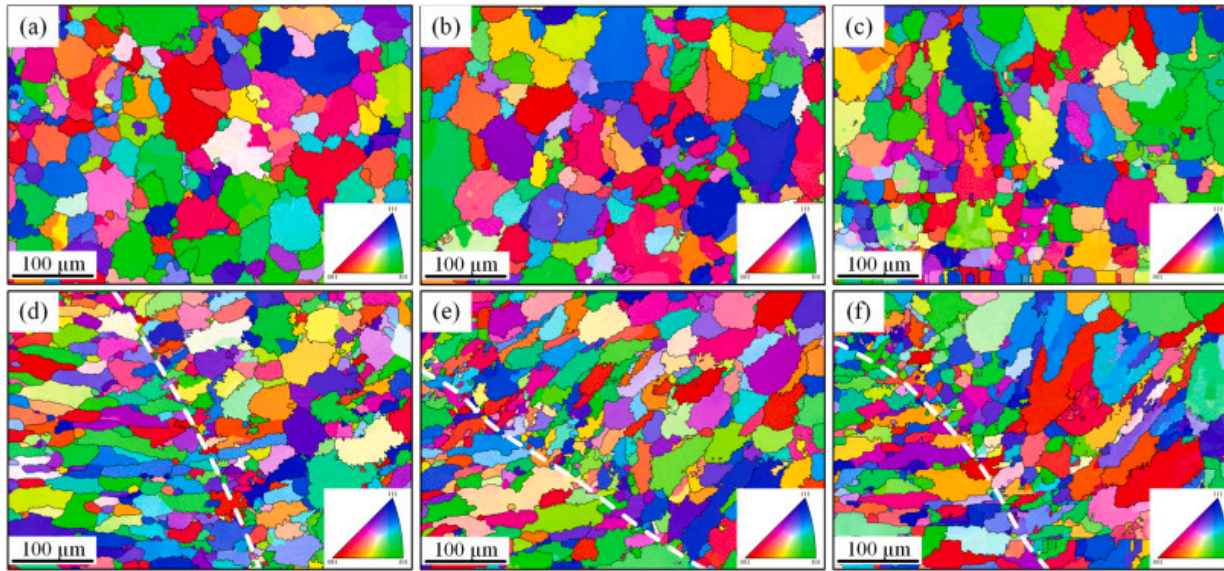


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Fig. 22. Variation of solidification parameters along the weld width direction of the cross-section under various ring power ratios (RPR). (a) Temperature gradient G ; (b) solidification rate R ; (c) cooling rate GR .

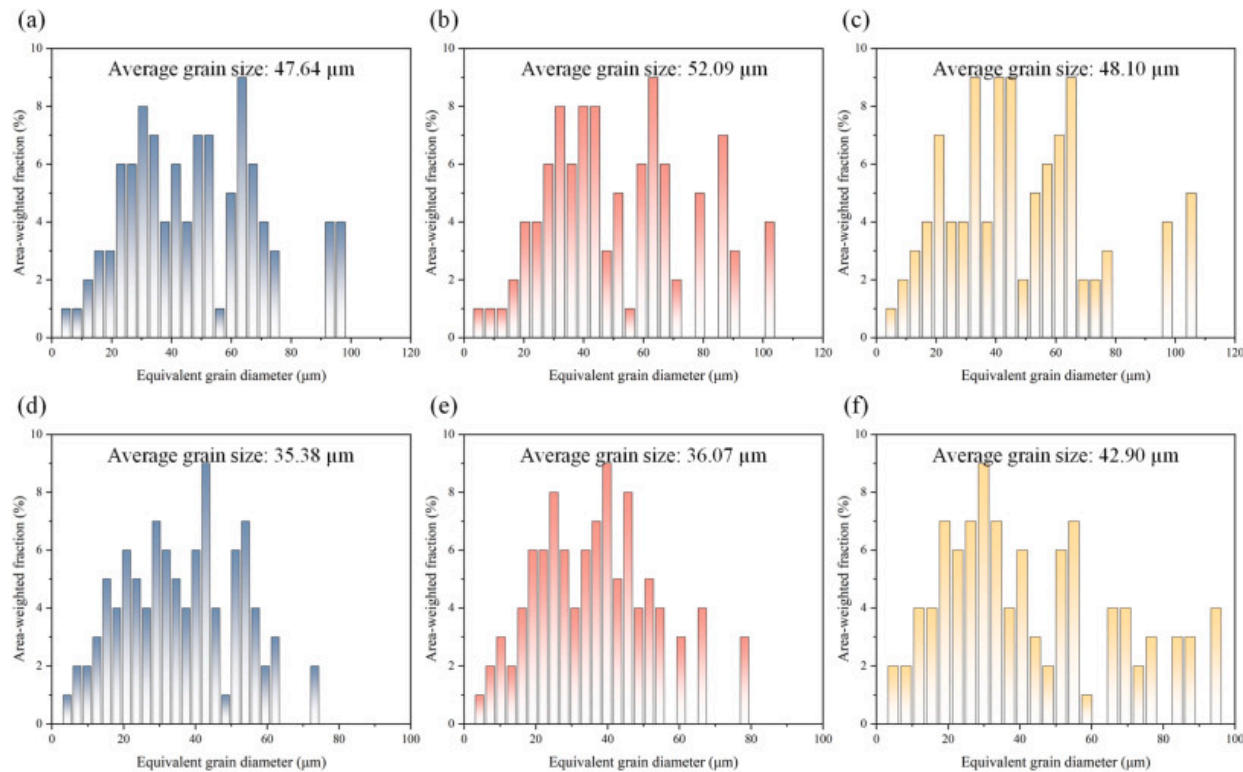
To verify the simulation results, we performed EBSD analysis on the center of the weld cross-section and the fusion boundary, extracted the grain morphology, and statistically analyzed the grain size at different locations. Fig. 23 demonstrates the microstructure at the center of the cross-section and at the fusion boundary for various RPR. The white dashed line represents the fusion boundary, and the region to the left of the fusion line is the base material. It was noticeable that the central fusion zone consisted entirely of equiaxed grains, with small equiaxed grain bands interspersed near the fusion boundary. The grains then grew vertically from the fusion line towards the center in a columnar structure, with the growth direction aligned with the temperature gradient. As indicated in Fig. 22(a), the temperature gradient G in the central fusion zone was substantially smaller than that on its sides. A lower G tended to promote the production of equiaxed grains, while a higher G favored the formation of columnar grains, which was consistent with the findings of [51]. Fig. 24 displays the distribution of grain size at different locations. In the central fusion zone, the minimum average grain size was $47.64\mu\text{m}$ ($\text{RPR}=0$), and the maximum average grain size was $52.09\mu\text{m}$ ($\text{RPR}=25\%$), as seen in Fig. 24(a) and (b). For grain sizes below $50\mu\text{m}$, the percentage of grains was 58% for $\text{RPR}=0$ and $\text{RPR}=50\%$, while it was only 52% for $\text{RPR}=25\%$. For grain sizes above $80\mu\text{m}$, the percentages were 8% and 9% for $\text{RPR}=0$ and $\text{RPR}=50\%$, respectively, while it was 14% for $\text{RPR}=25\%$. The higher percentage of larger grains for $\text{RPR}=25\%$ results in the largest average grain size. At the fusion boundary, the $\text{RPR}=50\%$ exhibited the largest average grain size ($42.90\mu\text{m}$), as shown in Fig. 24(f). No grains larger than $80\mu\text{m}$ were observed at the fusion boundaries for $\text{RPR}=0$ and $\text{RPR}=25\%$, whereas 10% of the grains exceeded this size for $\text{RPR}=50\%$ (as also seen in Fig. 23(f)), with larger columnar grains contributing to the larger average grain size. The grain size results at the cross-sectional center and the fusion line boundary were consistent with our previous predictions.



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Fig. 23. Grain morphology at different locations across the weld cross-section under various ring power ratios (RPR). (a–c) Grain morphology in the fusion zone with RPR=0, RPR=25%, and RPR=50%; (d–f) grain morphology at the fusion boundary with RPR=0, RPR=25%, and RPR=50%.



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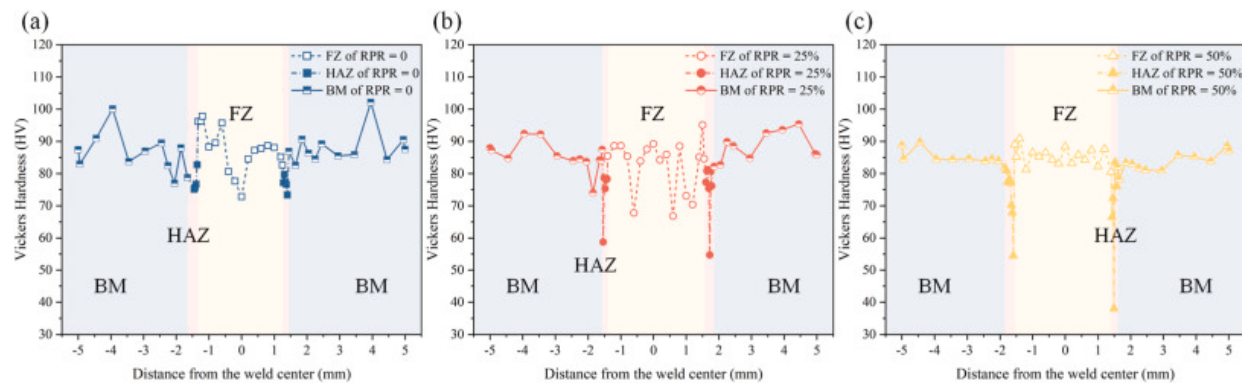
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Fig. 24. Grain size distribution at different locations under various ring power ratios (RPR). (a–c) Grain size in the fusion zone with RPR=0, RPR=25%, and RPR=50%; (d–f) grain size at the fusion boundary with RPR=0, RPR=25%, and RPR=50%.

3.5. Influence of ring power ratios on mechanical properties

To further investigate the effect of various RPR on mechanical properties, the Vickers hardness distribution was measured along the horizontal cross-section of the sample, as shown in Fig. 25. After that, a statistical analysis was performed on the hardness data at different positions for various RPR. According to the results, the average hardness value of the base material ranged from 80.6 HV to 86.3 HV. The average hardness values of the

fusion zone under various RPR were 87.9 HV (RPR=0), 82.9 HV (RPR=25%), and 85.4 HV (RPR=50%), which were comparable to the hardness level of the base material. The smallest and largest average grain sizes in the fusion zone were observed with RPR=0 and RPR=25%, respectively, leading to the highest and lowest average hardness values in the fusion zone. As depicted in Fig. 23(d–f), the width of the heat-affected zone at the fusion boundary increased with the rise of RPR. By decreasing the measurement spacing and the size of the hardness test points, we were able to obtain sufficient hardness data within the heat-affected zone, even though its width was still relatively narrow. The average hardness values at the fusion boundaries were 77.2 HV (RPR=0), 74.6 HV (RPR=25%), and 70.4 HV (RPR=50%). The decrease in average hardness was attributed to grain coarsening and the development of large columnar crystals, resulting from the lower cooling rate and overheating. The RPR=50% exhibited the most significant coarsening of columnar crystals (Fig. 23(f)), resulting in the smallest average grain size. Interestingly, some high hardness values were found at the fusion boundaries with both RPR=25% and RPR=50%. This may be because the broader zone of fine equiaxed grains at the fusion boundary increased the probability of these areas being sampled by the hardness tester.

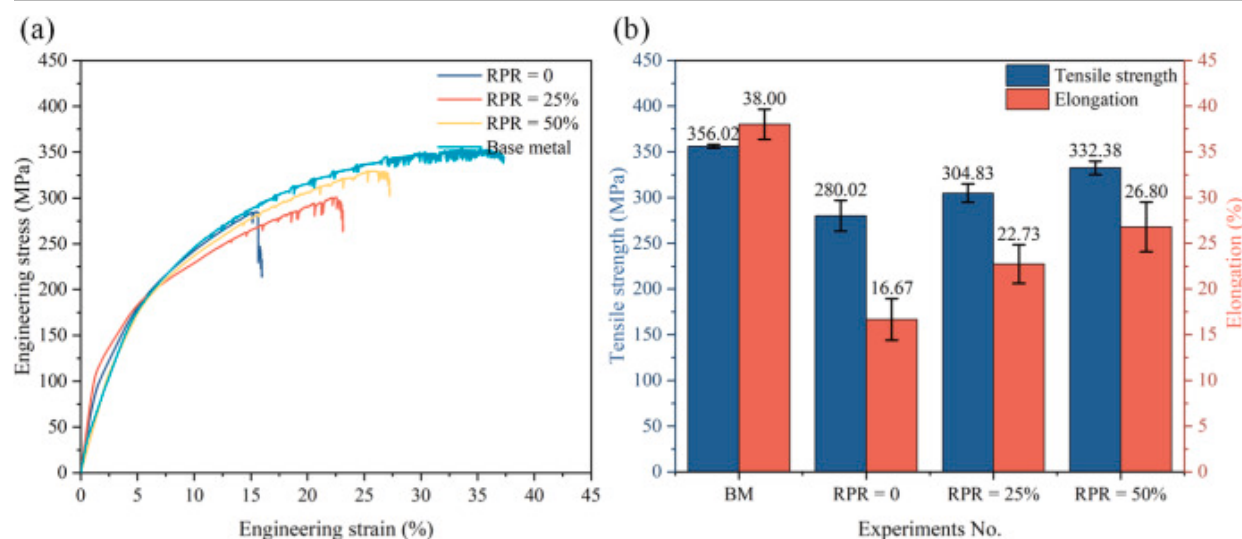


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Fig. 25. Microhardness distribution of the welded joint under various ring power ratios (RPR). (a) RPR=0; (b) RPR=25%; (c) RPR=50%.

In addition, we prepared three sets of test specimens for both the base material and each processing method, and conducted tensile tests to evaluate their mechanical strength. The corresponding stress-strain curves, as well as the average tensile strength and elongation, are shown in Fig. 26. The average tensile strength of the weld joints with various RPR ranged from 280.02 MPa to 332.38 MPa, which was approximately 78.7%–93.4% of the average tensile strength of the base material (356.02 MPa). The average elongation ranged from 16.67% to 26.80%, which is approximately 43.9%–70.5% of the average elongation of the base material (38%). Tensile strength is generally inversely proportional to grain size, smaller grain sizes can improve tensile strength to some extent [52], but this is under the condition of no significant defects. It can be observed that although the RPR=0 had the smallest grain size, the numerous internal pores (Fig. 5(a)) and the rough weld surface (with surface defects, as shown in Fig. 4(a–c)) led to stress concentration, promoting crack initiation and propagation, thus weakening the load-bearing capacity. Due to its flat weld surface and extremely low internal porosity, the RPR=50% demonstrated the highest tensile performance.

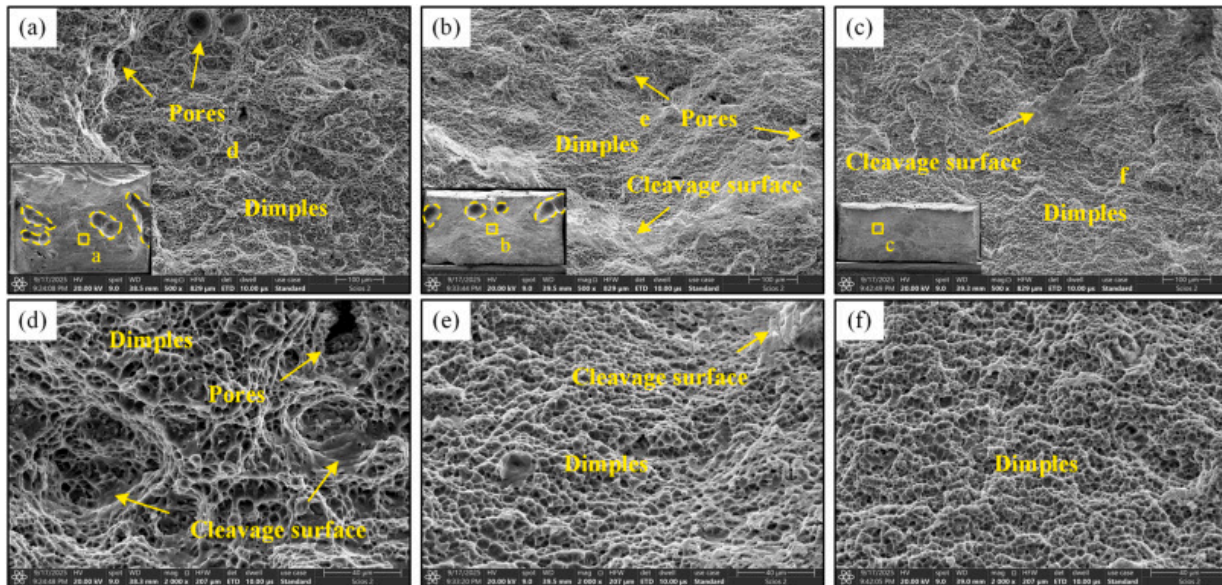


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Fig. 26. Tensile test data under various ring power ratios (RPR). (a) Stress-strain curves;
(b) average tensile strength and elongation.

The fracture surface morphologies of welded joints with various RPR are displayed in Fig. 27. The macroscopic fracture surface morphology exhibited that the joint of RPR=0 had numerous pore defects of varying sizes and the worst weld surface. The increase of RPR significantly reduced the number and size of pores, while the surface flatness was greatly enhanced, as illustrated in Fig. 27(a–c). The fracture surfaces at high magnifications of RPR=0 and RPR=25% both contained texture features such as pores, cleavage surfaces, and dimples, while no visible pores were found at RPR=50%. In addition, as the RPR increased, the RPR=25% and RPR=50% included more dimples and smaller dimple sizes within the same area, which all resulted in better tensile properties at high ring power ratios.



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Fig. 27. Fracture morphologies of weld joints with various ring power ratios (RPR). (a) and (d) are the macroscopic morphology and high magnification morphology with RPR=0; (b) and (e) are the macroscopic morphology and high magnification morphology with RPR=25%; (c) and (f) are the macroscopic morphology and high magnification morphology with RPR=50%.

4. Conclusion

In this study, a ring-mode laser was used for laser welding of aluminum alloys. The effects of various ring power ratios (RPR) on the weld bead morphology, internal porosity, and mechanical properties were investigated. Furthermore, a three-dimensional transient multi-physics model was developed to explore the mechanisms by which keyhole dynamics and the distribution of the heat flow in the molten pool during ring-mode laser welding (RMLW) were impacted by various RPR. The profile of the simulated weld fusion zone closely resembled the experimental results. Based on heat distribution and metal flow, the solidification characteristics of the molten pool under various RPR were predicted, and the accuracy of these predictions was verified by contrasting them with the experimentally determined weld microstructure. The following are the main conclusions of this study.

- 1) An increased RPR helped mitigate weld spatter, the RPR=50% reduced the height difference along and perpendicular to the weld surface by 71.5% and 71.6%, respectively, resulting in a more uniform surface texture and superior surface flatness. Simultaneously, it significantly reduced the number and size of pores, the RPR=50% resulted in an 80.8% reduction in porosity. The number of pores initially declined, then rose again, and ultimately fell to a minimum value due to the reduction in fusion area.
- 2) A more regular flow pattern with longer flow paths and fewer eddies within the molten pool was produced by increasing the RPR. This reduced the influence on the middle keyhole and top entrance, effectively preventing

expansion and contraction of the molten pool, and mitigating keyhole oscillations. Furthermore, the wider molten pool and longer upward flow enhanced the probability of bubbles escaping to the surface.

- 3) Increasing the RPR caused the laser energy to shift towards the keyhole bottom, significantly reducing the maximum temperature difference at different positions along the penetration, resulting in a more homogeneous energy distribution on the keyhole wall. G and GR were significantly decreased.
- 4) Even without a smaller grain size, the increased RPR displayed superior tensile strength because of better weld surface flatness and fewer internal porosity defects. Tensile strength increased by 14.7%.

The influence of various RPR on the molten pool's solidification characteristics and the heat flow distribution during RMLW of aluminum alloys was examined in this work. Based on the solidification characteristics, the feasibility of controlling the final solidification microstructure by adjusting the RPR is verified, thus providing an efficient method for tailoring the microstructure of aluminum alloy welds in RMLW. However, it is worth noting that this study has certain limitations: the limited computational resources resulted in insufficient simulation time, and the relatively low welding speed led to a negligible molten pool tail. It is impossible to predict the width of the equiaxed grains with accuracy due to these considerations, which causes the solidification parameter along the weld centerline to be significantly greater than 0. We will further improve the model in future research to overcome these limitations.

CRediT authorship contribution statement

Kai Guo: Writing – original draft, Visualization, Methodology, Investigation. **Fengye Tang:** Validation, Methodology. **Honghe Jia:** Resources. **Wenyong Zhao:** Supervision, Resources,

Conceptualization. **Yanhong Wei**: Writing – review & editing, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors gratefully acknowledged projects supported by the General Program of the National Natural Science Foundation of China (Grant No. [52275341](#)), the National Natural Science Foundation of China (Grant No. [52405366](#)).

Data availability

Data will be made available on request.

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